

NEURAL BELLMAN OPERATORS

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We develop Neural Bellman Operators for continuous-time economic decisions. A detached continuation-evaluation step supplies marginal values to a feasible policy-improvement step. We establish evaluation, policy-error, and continuous-economy transfer results, and derive a signed occupation Bellman account that preserves cancellation between predicted improvement and continuation error. A common-continuation result identifies which observations can be reused across current tasks. For an independently fixed scalar-gradient dictionary, an exact decision-value identity characterizes the economic return to learning through removed noise variance and approximation cost. In a capital economy, independent signed assessments establish positive continuous-economy gains in dimensions ten and fifty. A prospectively specified five-method comparison gives all methods legitimate reuse. NBO improves materially on fitted Raw actors in difficult repeated-decision cases and achieves practical payoff equivalence with cached sample-average optimization at lower measured construction and query work in six of eight final-stage cases. The sample-average optimizer attains higher payoffs in the two long-continuation cases. Retraining an exact-trace HJB family across four economic designs substantially narrows the original comparator gap. Whole-segment scalar guarantees provide absolute decision-class accuracy without being identified with high-dimensional near-optimality. The theory, economic applications, and complete favorable and adverse comparisons distinguish learned continuation, feasible improvement, computational cost, and solution accuracy.

KEYWORDS: Dynamic programming, Neural approximation, Recursive utility, Endogenous preferences, Viscosity solutions, Dynamic games.

1. INTRODUCTION

An economic decision maker often changes a current choice while leaving its continuation problem unchanged. A tax or consumption constraint may change today, a portfolio manager may face a new current mandate, or an optimizer may revisit a nearby state. The continuation value is a common economic object across these decisions. Learning it can replace repeated continuation calculations by evaluations of one function. The relevant question is not whether the function interpolates its training observations, but whether its marginal values improve feasible decisions enough to justify learning and deploying it.

This paper develops Neural Bellman Operators (NBO) around that question. The operator separates policy evaluation from policy improvement. A continuation representation is fitted while the reference policy is held fixed. The actor then uses detached continuation channels to choose a feasible policy. The distinction matters both analytically and computationally: a joint derivative through the actor and its moving evaluation target is not the same map. A final independent economic comparison assesses the returned policy. The framework applies to continuous-time control and, under their additional assumptions, recursive utility, endogenous preference formation, temporal selves, and dynamic games. These applications and their original derivations remain in the current supplement; the quantitative method comparisons concern the capital economy developed in the article.

The first contribution is an error account from a learned continuation to a feasible policy and its continuous-economy payoff. The analysis distinguishes a fixed-policy numerical continuation from the limiting Bellman solution, and a held-action controller from a continuously updated feedback. The feasible analytical schedule supplies an explicit economic anchor. Candidate-occupation error, maximization error, holding loss, arithmetic, and diffusion transfer enter different terms. The general evaluation, viscosity, and improvement results are stated with the assumptions that support them, rather than inferred from a small training residual.

The second contribution is a signed occupation account. A norm-based sufficient condition replaces the continuation correction by a worst-case magnitude and can lose the cancellation that actually occurs along the policy's action changes. We

1 instead evaluate common-noise continuation differences and form predicted im- 1
 2 provement minus the signed correction before statistical inference. The expectation 2
 3 is exactly the finite Bellman gain. An antithetic range calculation then transfers a 3
 4 fresh lower endpoint to the continuous economy. In the executed assessment, the 4
 5 original negative norm-based bounds coexist with positive signed lower bounds 5
 6 of at least 0.000767 and 0.000776 in dimensions ten and fifty. This is evidence 6
 7 that the returned NBO policies realize positive occupation Bellman gains, not an 7
 8 identification of the critic’s unique incremental contribution relative to Raw. 8

9 The third contribution concerns the economic value of the common continuation 9
 10 itself. We state precisely when a rollout, its innovations, and its derivative can be 10
 11 reused. For a scalar-gradient dictionary fixed independently of the fitting labels, the 11
 12 reduction in prediction risk is exactly removed noise variance less approximation 12
 13 cost. A quadratic-decision specialization translates the same decomposition into a 13
 14 payoff comparison. These results exhibit a positive economic mechanism for learn- 14
 15 ing, including strict improvement when the dictionary is correctly specified and 15
 16 removes relevant noise. They also show why the mechanism cannot be assumed: 16
 17 a misspecified representation may cost more than the variance it removes, and a 17
 18 Raw competitor using the same projection receives the same benefit. The completed 18
 19 continuation-menu comparison gives every method its legitimate reuse rather than 19
 20 forcing Raw to regenerate observations unnecessarily. Across four finite-period 20
 21 designs, two dimensions, and three fixed work stages, it retains 128 trials and 640 21
 22 method fits. At the last stage, the long fifty-dimensional case has material NBO 22
 23 payoff advantages over the fitted vector costate, Raw actor, and direct actor. In six 23
 24 of the eight final-stage economic–dimension cases, the entire NBO-minus-cached- 24
 25 sample-average interval lies inside the prespecified practical-equivalence band, 25
 26 while NBO uses less construction and query work. In both long-continuation cases, 26
 27 the cached sample-average optimizer has a material payoff advantage. Thus the ex- 27
 28 periment identifies an economic work–accuracy tradeoff, not universal dominance 28
 29 by a learned representation. 29

30 The numerical evidence makes three distinctions essential to the economic in- 30
 31 terpretation. First, independent confirmation of a fixed policy is different from 31
 32 retraining a method on new streams. A new prospective bank confirms the original 32

NBO–HJB paired comparison in both dimensions. A separate completed experiment retrains the exact-trace HJB family over four economic designs and eight new streams per design and dimension. The strengthened comparator substantially narrows the original gap: the low-volatility fifty-dimensional contrasts remain positive against both deployments, whereas the other fourteen HJB contrasts are unresolved. Second, an interval containing zero is not evidence of equivalence or noninferiority. The original NBO–Raw and NBO–direct-policy comparisons remain unresolved after fresh confirmation; Raw remains cheaper in the original work account. Third, positive improvement over the analytical schedule is not a tight solution of the high-dimensional HJB problem. The broad adapted-control gap bounds, classical scalar comparisons, and state-specific ranking reversals are retained and interpreted at their stated scope.

The article presents the operator, the capital economy, the signed mechanism, and the economic evidence. The supplement contains the full theorem chain, proofs, original applications, complete adverse and favorable numerical accounts, and the prospectively fixed strengthened-HJB and continuation-menu designs together with every registered outcome. A separate response identifies the revision corresponding to each referee comment. Throughout, economic primitives, training-stream distributions, selected policies, and confirmation families remain distinct objects.

1.1. *Relation to the literature*

Neural residual methods build on numerical dynamic programming and function approximation, not on a new principle of optimality. Projection and related methods exploit economic structure to approximate a Bellman equation; see [Judd \(1998\)](#). Derivative approximation by smooth neural networks provides useful representation results ([Hornik et al., 1990](#)), but does not establish that a particular finite architecture or optimizer solves a nonlinear boundary-value problem.

The differential critic is closely related to the neural PDE approach in [Sirignano and Spiliopoulos \(2018\)](#). Their DGM formulation includes differential and boundary conditions and treats HJB equations. The deep BSDE method of [Han et al. \(2018\)](#) also includes an HJB application. The distinguishing object here is the combination of

an explicit feasible actor, block-specific evaluation, and economic error accounting across control and equilibrium problems. An analytical formula for the maximizing action is helpful when available, but is not a universal dividing line between neural PDE methods. Likewise, recursive dependence on the value is a nonlinear zeroth-order term; full nonlinearity in the PDE sense concerns the highest derivatives.

Recursive utility separates intertemporal substitution from risk aversion (Epstein and Zin, 1989, Duffie and Epstein, 1992). Its infinite-horizon formulation requires particular care about utility-process selection and transversality; Herdegen et al. (2021) explains why a stationary algebraic solution alone is insufficient. The time-inconsistency extension is an equilibrium problem in the sense of Björk et al. (2016), not an ordinary control problem with a relabelled flow payoff. Finally, stochastic approximation results require their own step-size, stability, and limiting-dynamics conditions (Borkar, 2024). They do not convert constant-step Adam into a convergence theorem for economic optimality.

The article proceeds from the economic problem to its evaluation and improvement map. Section A states the general verification results. Sections 4–A.2 specialize the welfare account to the capital economy. Section B states the other economic applications. The computational study distinguishes the existing comparison from the current experiment. The supplement collects the complete models, derivations, proofs, and supporting diagnostics by subject.

2. THE CONTROLLED ECONOMY

Let W be an m -dimensional Brownian motion on a filtered probability space satisfying the usual conditions. The state $S = (u, X)$ contains internal and external state variables. An action $a = (\theta, c, \pi, \dots)$ combines internal adjustment and external choices. The controlled state evolves as

$$dS_s = b(s, S_s, a_s) ds + \sigma(s, S_s, a_s) dW_s, \quad a_s \in \mathcal{A}(s, S_s). \quad (1)$$

The compact feasible correspondence $\mathcal{A}$ is part of the model. Actions are progressively measurable and must generate a well-defined state process satisfying the declared state restrictions. State viability does not follow from bounded actor outputs alone.

We work initially on a finite horizon T . A computational domain D is either invariant, equipped with a specified reflection mechanism, or stopped on exit. In the stopped case let $\tau = T \wedge \inf\{s \geq t : S_s \notin D\}$. The payoff $g(\tau, S_\tau)$ is specified on the entire parabolic stopping boundary, including the terminal face. For a fixed admissible policy a , utility is the policy-specific solution

$$Y_s^a = g(\tau, S_\tau) + \int_s^\tau f(v, S_v, a_v, Y_v^a) dv - \int_s^\tau Z_v^a dW_v. \quad (2)$$

Adjustment costs are included in f . For example, time-additive utility has $f(s, S, a, y) = r_0(s, S, a) - \rho y$, where r_0 includes all current resource and adjustment costs. Define $V^a(t, x) = Y_t^{a, t, x}$ and $V^*(t, x) = \sup_a V^a(t, x)$. The value inside a candidate policy's recursive objective is its own utility process, not the unknown optimum inserted into every candidate objective.

ASSUMPTION 1—Finite-horizon comparison class: *The controlled state problem is well posed for the admissible policies used below. The coefficients are continuous, locally Lipschitz in the state uniformly over actions, and bounded on the compact stopped domain. The payoff is continuous and bounded. On an invariant utility interval containing all policy values, f is continuous in its arguments and uniformly Lipschitz in utility with constant $L \geq 0$. The admissible class is stable under concatenation and has the measurable-selection properties needed for dynamic programming. The resulting HJB problem and the fixed-policy problems obey comparison in the bounded continuous class with the specified boundary conditions.*

The comparison requirement identifies both a function class and a boundary problem. It is not asserted merely because the primitive coefficients are continuous. On a bounded, regular Dirichlet domain, a boundary barrier is one way to verify attainment of the stopping payoff. Reflection requires the corresponding Neumann or oblique derivative condition and a well-posed reflected state process. Unbounded-state applications require growth bounds and localization. These conditions are checked separately from a neural residual.

For a smooth function v , write

$$\mathcal{L}^a v = b(t, x, a)^\top Dv + \frac{1}{2} \text{tr}\{\sigma(t, x, a)\sigma(t, x, a)^\top D^2 v\}, \quad (3)$$

$$\mathcal{H}^a[v] = f(t, x, a, v) + \mathcal{L}^a v, \quad F[v] = -v_t - \sup_{a \in \mathcal{A}(t, x)} \mathcal{H}^a[v]. \quad (4)$$

The value solves $F[V^*] = 0$ with the specified boundary data, in the viscosity sense where classical derivatives do not exist. The diffusion covariance is the positive semidefinite matrix $\sigma\sigma^\top$. No claim that the product of two arbitrary positive semidefinite matrices is symmetric positive semidefinite is needed.

A continuous martingale representation can also be used when its quadratic variation is Markov in an augmented state. If $d\langle M \rangle_s = Q(s, S_s, a_s)ds$, the covariance in (3) becomes $\sigma Q\sigma^\top$. A merely predictable process Q_s is not, without further state augmentation, a deterministic coefficient of a Markov HJB equation. Quadratic variation by itself also does not imply conditionally Gaussian increments. The implementation in this paper uses the Brownian model (1).

2.1. Infinite horizon and economic normalization

For a stationary problem, the boundary-value equation is $\sup_a \{f(x, a, V) + \mathcal{L}^a V\} = 0$. We use an infinite-horizon result only after specifying admissibility, integrability, and the selected utility process. In the time-additive case, a sufficient verification requirement is that the localized Itô identities are uniformly integrable and $\mathbb{E}[e^{-\rho T} | V(S_T) |] \rightarrow 0$ for the policies being compared. Recursive problems require an appropriate utility comparison and transversality condition instead of simply borrowing the additive one.

Cardinal transformations also require care. Multiplying a fixed utility function by a positive constant does not change its maximizer. Adding a function of an *endogenously controlled preference state* generally does. Consumption units, preference normalization, and terminal utility are therefore primitive economic choices in Section I.3.

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The actor $a_\phi(t, x)$ and critic $v_\xi(t, x)$ represent policies and continuation values. An internal actor and an external actor can be separate parameter blocks; their joint action remains the input to the same economic generator. A smooth activation is

useful for exact differentiation, while a nonsmooth neural representation can be used with the positive-weight Bellman formulation below.

3.1. Feasibility and boundary data

For a box $[a, \bar{a}]$, a sigmoid map produces interior feasible actions but does not attain its endpoints at finite weights. Binding constraints can instead be represented by a projection onto the closed interval, by an explicit boundary candidate, or by a finite action set including the endpoints. The constrained portfolio calculation uses a projection and can attain a zero risky share exactly. On a simplex, a probability map supplies a feasible interior point and boundary faces must again be considered when certifying the optimum. General constraints require a feasible parameterization, a certified projection, or a constrained action solver. A finite penalty is not an exact feasibility certificate.

On a finite horizon, an elementary hard-terminal architecture is

$$v_{\xi}(t, x) = G(x) + (T - t)N_{\xi}(t, x). \quad (5)$$

This enforces the terminal face, but not automatically a lateral boundary. A compatible lifting G_B and a factor d_B vanishing on all Dirichlet faces give $v_{\xi} = G_B + d_B N_{\xi}$. Alternatively, boundary residuals are trained separately and retained explicitly in the error bound. A reflected preference state requires its own derivative condition; a terminal architecture does not enforce it. For recursive utility, a domain-preserving transformation must additionally keep the continuation value in the domain of the aggregator.

3.2. The block-specific computational graph

Let ν be a specified distribution of state–time points and let sg denote stop-gradient. A critic step minimizes

$$L_C(\xi; \bar{\phi}) = \mathbb{E}_{\nu} \left[-\partial_t v_{\xi} - \mathcal{H}^{\text{sg}(a_{\bar{\phi}})}[v_{\xi}] \right]^2 + L_B(\xi), \quad (6)$$

where L_B is absent only for boundary conditions enforced exactly. The policy, sampling locations, and any rival actions are detached. The value, time derivative, state gradient, and Hessian remain differentiable with respect to the critic parameters.

An actor step minimizes

$$L_A(\phi; \bar{\xi}) = -\mathbb{E}_\nu \mathcal{H}^{a\phi}[\text{sg}(v_{\bar{\xi}}), \text{sg}(Dv_{\bar{\xi}}), \text{sg}(D^2v_{\bar{\xi}})]. \quad (7)$$

All critic derivative channels are fixed in this step. There is no critic gradient from L_A and no actor gradient from L_C . A printed sum $L_A + \omega L_C$ may be a reporting device, but it is not jointly differentiated. The implementation exposes separate functions and regression tests for these derivative paths.

The distinction is substantive: joint differentiation can move the critic away from the policy-evaluation root. Appendix C gives a counterexample, the recursive lifetime policy derivative, and the conditioning requirements. The local Hamiltonian objective is not identified with a lifetime policy gradient under arbitrary state sampling. The evaluated reference policy may remain fixed while a separate candidate actor is improved at successive work levels. Updating that reference defines a new evaluation target and is recorded explicitly.

3.3. Rollout and costate evaluation

For a frozen policy $a^{(j)}$, a positive-time Euler rollout from (t, y) supplies a sample return $\widehat{V}^{(j)}(t, y)$ and its pathwise derivative $\widehat{p}^{(j)}(t, y)$. Differentiation includes the state dependence of subsequent policy actions. Freezing the policy means freezing its parameters, not deleting that derivative. The targets evaluate the current policy, not the optimal policy, and contain no maximizing action labels. For fixed finite weights, bounded actions and smooth bounded production justify pathwise differentiation of this finite rollout away from the harmless clipping kinks. These are derivatives of the discretized evaluation problem, not asserted unbiased derivatives of its continuous-time limit.

The critic minimizes

$$\widehat{\mathbb{E}} \left[0.05 \{v_\theta - \widehat{V}^{(j)}\}^2 + \|d(D_y v_\theta - \widehat{p}^{(j)})\|_d^2 \right]. \quad (8)$$

The factor d puts normalized marginal values on an order-one scale. The rollout implementation uses two hidden tanh layers and a hard terminal component. Holding its new derivative fixed, the actor then maximizes

$$\hat{\mathbb{E}} \left[\overline{\log a_\phi} - \frac{\zeta}{2} \bar{a}_\phi^2 - a_\phi \cdot \text{sg}(D_y v_\theta) \right]. \quad (9)$$

Terms independent of actions have been omitted. Consumption is represented directly by $m_0(t) + \epsilon \tanh N_\phi(t, y)$; there is no online Hamiltonian search in its deployment. Value evaluation, costate evaluation, and feasible improvement are distinct operations, with their own stored losses and work counts. The number of critic and actor steps per rollout batch is a declared design parameter. Failure of either numerical block is recorded, not interpreted as successful policy improvement.

The stochastic gradient targets make the economically relevant marginal continuation values visible to the evaluation loss. This is not a new Bellman principle, nor an assumption that derivative regression invariably dominates differential residual fitting. It is a concrete implementation of the NBO evaluation map. The preserved differential implementation and its two independent trace banks continue to test a different evaluation procedure. Both feed the same economic distinction between policy evaluation and feasible improvement.

3.4. The implemented map and its verification output

Fix the economy, boundary realization, network classes, optimization budget, and verification procedure, denoted collectively by ϑ . Let $\mathcal{P}$ be the admissible feedback policies, Ξ the critic parameter space, and Ω the sampling streams. The implemented NBO is the numerical map

$$\mathcal{N}_\vartheta : \mathcal{P} \times \Xi \times \Omega \longrightarrow \mathcal{V}_\vartheta \times \mathcal{P}_\vartheta \times \mathcal{C}_\vartheta, \quad (a, \xi, \omega) \longmapsto (v, \tilde{a}, C). \quad (10)$$

The first component is approximate policy evaluation; the second is feasible improvement; the third records verification of the resulting pair. A record identifies the model, parameters, domain, boundary faces, evaluation error, maximization error, coverage, discretization errors, and unchecked assumptions. An unavailable bound is not zero. Failure to obtain a useful certificate is a possible output of the

map. With its sampling stream fixed, the implementation is deterministic. We do not assert that a network learns an unrestricted Bellman operator on a function space.

The evaluation–improvement structure is an instance of approximate policy iteration, including its partial-evaluation variants (Scherrer et al., 2015). Neural representation does not introduce a new principle of optimality, and monotonicity belongs to the underlying positive-weight scheme, not to gradient descent. The NBO specification joins the block-specific derivative graph to boundary and utility-domain restrictions and to a common economic error account. Sections M.1 and I.3.4 implement the differential and positive-weight versions of this map, respectively, with independent verification of their outputs.

3.5. The theorem chain and economic scope

The general evaluation and improvement results are collected in the supplement, with their complete statements and proofs retained. Fixed-policy evaluation, the selection of a Bellman solution, and the performance of the returned actor are separate claims. Their hypotheses specify admissibility, comparison, regularity or monotone approximation, and the relevant terminal or transversality condition. Recursive utility, endogenous preferences, temporal selves, and dynamic games retain their own hypotheses; the capital experiment does not provide numerical validation for those distinct applications.

The capital development below specializes this chain to explicit feasible actions, a tractable schedule, and a payoff statistic in economic units. It then separates two questions. The first is whether the selected policy produces a positive gain. The second is whether learning the continuation improves decisions or work relative to a credible procedure without that scalar critic. The signed occupation account addresses the first question. The common-continuation and decision-risk results describe a mechanism for the second, which still requires an appropriate empirical comparison for the implemented nonlinear methods.

Algorithm 1 Neural Bellman evaluation, feasible improvement, and economic assessment

Require: Economic primitives and information; feasible actions and utility domain; boundary data; evaluation variant; training, validation and stopping rules; independent final assessment.

- 1: Declare the feasible reference policy to be evaluated. Initialize a feasible candidate and a boundary-compatible continuation representation.
 - 2: **for** each permitted evaluation-improvement iteration **do**
 - 3: Freeze the declared reference policy, rivals and evaluation target. Fit only the critic by the declared differential, positive-weight, or rollout rule. The reference may remain fixed across work levels.
 - 4: Freeze the critic and its value/derivative channels. Update only the feasible actor, or apply the declared feasible action solver.
 - 5: Record all evaluation and improvement work, corrections, unsuccessful checks, and the resulting deployment rule.
 - 6: Apply the predetermined validation and stopping rule, using no final-test outcomes.
 - 7: **end for**
 - 8: Freeze the selected policy and continuation object before independent final evaluation.
 - 9: Compute the specified economic endpoints and direct contrasts, with their approximation, implementation and statistical accounts.
 - 10: Return the fitted objects, their implementation and domain, and the complete accuracy and work record.
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4. CONTINUOUS-ECONOMY GUARANTEES FOR NEURAL CAPITAL POLICIES

The distinction between a nodal certificate and an economic diffusion bound is substantive. This section obtains the latter for the original dense, nonlinear capital economy, without replacing its primitives, restricting the comparison policy class, or extrapolating a sampled residual. The construction uses the curvature of consumption utility to control the gain from an arbitrary adapted deviation. A neural actor is then confined to a known neighbourhood of a feasible schedule.

Its approximation class is restricted; the economic optimum against which it is assessed is not.

Write $\bar{z} = d^{-1} \sum_i z_i$, $\|z\|_d^2 = d^{-1} \sum_i z_i^2$, and $\Pi z = z - \bar{z} \mathbf{1}$. In the capital model already specified, the state and payoff are

$$dY_i = \{a + \kappa \tanh((BY)_i) - m_i - v^2/2\} dt + \sigma_I dW_i + \sigma_C dW_0, \quad (11)$$

$$J_y(m) = \mathbb{E}_y \left[\int_0^T e^{-\rho t} \{\overline{\log m_t} + \bar{Y}_t - \frac{\zeta}{2} \bar{m}_t^2\} dt + e^{-\rho T} \{\bar{Y}_T - \chi \|\Pi Y_T\|_d^2\} \right], \quad (12)$$

where $v^2 = \sigma_I^2 + \sigma_C^2$ and each control lies in $[\underline{m}, M]$. Controls are progressively measurable and the state equation is interpreted on $\mathbb{R}^d$. Bounded control and the globally Lipschitz bounded production term yield the required state moments and finite payoffs. The historical parameters are $T = 1$, $\rho = 0.04$, $a = 0.10$, $\kappa = 0.15$, $\sigma_I = 0.15$, $\sigma_C = 0.10$, $\zeta = 0.2$, $\chi = 0.03$, and $[\underline{m}, M] = [0.02, 2]$. The stored coupling matrices are unchanged in dimensions ten and twenty. Dimension fifty uses the same stated matrix-generation rule.

Define

$$w(t) = \frac{1 - e^{-\rho(T-t)}}{\rho} + e^{-\rho(T-t)}, \quad m_0(t) = \frac{2}{w(t) + \sqrt{w(t)^2 + 4\zeta}}. \quad (13)$$

This schedule is strictly interior for the parameters above. It is a feasible benchmark, not an asserted solution of the coupled economy. For the analytical comparison, keep m_0 interior and assume $\rho > 0$, $\zeta, \chi, \kappa \geq 0$, and let $\beta \geq \kappa \|B\|_2$ and $s_0 \geq \|\Pi y\|_d$. Choose $\epsilon \geq 0$ so that $[m_0(t) - \epsilon, m_0(t) + \epsilon] \subset [\underline{m}, M]$ at every date. With $V^*(y) = \sup_m J_y(m)$ over the full adapted admissible class, the schedule gap obeys $0 \leq V^*(y) - J_y(m_0) \leq G_d$. The exact scalar integral defining G_d is (S.35); Appendix E gives its curvature and dispersion constants, Theorem S.10, and the complete proof. This analytical bound supplies a common economic reference. The policy-specific endpoint below measures what the fitted actor adds to that reference.

4.1. *The deployed neural map and its numerical guard*

The actor is a two-hidden-layer network with the closed-form output map

$$m_i^\phi(t, Y) = m_0(t) + \epsilon \tanh N_{\phi,i}(t, Y). \quad (14)$$

The weights are trained by the same block-specific NBO losses as before. No maximizing policy labels, nodal action table, or online local search enter this actor. Input clipping at very large log-capital values protects numerical evaluation; it does not stop or truncate the economic state. The theorem remains valid for any bounded proposal followed by the stated output constraint, including this clipped-input network.

Numerical implementation must not replace the exact schedule by an unverified floating-point centre. The deployment routine therefore encloses $m_0(t)$ by $[\ell(t), u(t)]$ using the inherited outward arithmetic kernel and projects the proposal onto

$$[\text{up}(u(t) - \epsilon), \text{down}(\ell(t) + \epsilon)]. \quad (15)$$

This interval lies inside the exact tube. Adjacent clock floats enclose rounding of the input time. An empty interval causes an error; a nonfinite proposal is replaced by its interval midpoint and is not evidence of successful training. This scalar feasibility guard is neither an action-maximization oracle nor a stored policy correction table. Its cost is included in the deployment measurements.

4.2. *Policy-Specific Verification after Bellman Improvement*

The bound in Theorem S.10 is a useful starting point, but not a measure of what training accomplishes. Every policy in the same tube receives the same allowance, including a poorly trained policy. We now replace that allowance by an independently verified, policy-specific payoff difference. The economic primitives, initial state, and full adapted comparison class in (11)–(12) are unchanged. The implementation to which the new bound applies is specified explicitly below. It is not identified with continuously observed-state feedback.

4.2.1. The deployed controller

Fix a deterministic grid $t_k = kh$, $Nh = T$. The controller has an internal state Z_k , initially y , and observes the Brownian innovations at the grid dates. At date t_k it selects

$$a_k = \mathcal{P}_k\{m_\phi(t_k, Z_k)\}, \quad Z_{k+1} = Z_k + h\{c\mathbf{1} + f(Z_k) - a_k\} + \Sigma\Delta\tilde{W}_k, \quad (16)$$

where $c = a - \nu^2/2$, $f(z) = \kappa \tanh(Bz)$, and $\mathcal{P}_k$ projects inward onto the declared action interval around $m_0(t_k)$. The vector $\Delta\tilde{W}_k$ clips each standardized innovation at ten before rescaling by $\sqrt{h}$. The numerical implementation and its rounding convention form part of the controller. Between dates it holds a_k fixed. Its economic state, denoted Y^ϕ , nevertheless satisfies

$$dY_t^\phi = \{c\mathbf{1} + f(Y_t^\phi) - a_k\} dt + \Sigma dW_t, \quad t_k \leq t < t_{k+1}.$$

In particular, the economic Brownian motion and state are not clipped, stopped, or replaced by Z . The controller is adapted to the original innovation filtration and remains feasible. Observing innovations is an implementation assumption, not a conclusion inferred from a state-only observation model. The continuously evaluated absolute-bound study policies remain separate objects in the supplement.

This specification has an important analytical consequence. When the numerical state and the economic state are compared, they use exactly the same chosen actions. Their state discrepancy therefore depends on the smoothness of the economic drift, not on a global Lipschitz bound for the learned network. The following construction uses this fact rather than inserting sampled residuals into a uniform verification theorem.

4.2.2. An exact improvement identity

Write $W(t) = e^{-\rho t}w(t)$ and define the nonnegative consumption deficit

$$R_t(m) = \log m_0(t) - \overline{\log m} + w(t)\{\bar{m} - m_0(t)\} + \frac{\zeta}{2}\{\bar{m}^2 - m_0(t)^2\}. \quad (17)$$

The first-order identity $m_0^{-1} = w + \zeta m_0$ and concavity of the logarithm imply $R_t(m) \geq 0$. The exact mean-state identity in the proof of Theorem S.10 gives

$$\Delta_\phi := J_y(a^\phi) - J_y(m_0) = \mathbb{E} \int_0^T W(t) \overline{f(Y_t^\phi) - f(Y_t^0)} dt - \mathbb{E} \int_0^T e^{-\rho t} R_t(a_t^\phi) dt - \chi e^{-\rho T} \mathbb{E} \{ \|\Pi Y_T^\phi\|_d^2 - \|\Pi Y_T^0\|_d^2 \}. \quad (18)$$

Thus state-dependent consumption has to earn its utility-curvature cost through production or terminal dispersion. The identity is also an effective variance reduction: the common mean-state martingale cancels before simulation statistics are formed. Positive neural output is not itself evidence of positive economic improvement.

For each cell, calculate the scalar integrals

$$\begin{aligned} A_k &= \int_{t_k}^{t_{k+1}} e^{-\rho t} dt, & B_k &= \int_{t_k}^{t_{k+1}} W(t) dt, \\ C_k &= \int_{t_k}^{t_{k+1}} e^{-\rho t} \{ \log m_0(t) - w(t)m_0(t) - \zeta m_0(t)^2/2 \} dt. \end{aligned}$$

The consumption-deficit integral for a held action is exactly $C_k - A_k \overline{\log a_k} + B_k \bar{a}_k + \zeta A_k \bar{a}_k^2/2$. Consequently, consumption need not be approximated by a left-point utility sum. Let $X_{\phi,h}$ be the paired statistic obtained from (18) by using the internal Euler states in the production and terminal terms and these integrated consumption terms. The reference internal state uses the integrated common action $\int_{t_k}^{t_{k+1}} m_0(t) dt$. All scalar integral enclosures and their midpoint errors are saved.

4.2.3. Transferring the computed payoff to the economy

The statistic $X_{\phi,h}$ is evaluated for the held controller in (16), using the same actions when comparing its internal state with its economic diffusion. Proposition S.11 in Appendix E bounds $|\Delta_\phi - \mathbb{E} X_{\phi,h}|$ by the saved allowance $E_{\phi,h}^{\text{all}}$. The exact-arithmetic diffusion allowance is $E_\epsilon(h) + E_0(h)$, with E_ϵ defined in (S.44). The appendix gives the additional clipped-innovation, forward-arithmetic, and enclosed-integration terms. The $O(h)$ diffusion bound uses bounded actions, the global drift and derivative bounds of the specified hyperbolic-tangent production function, and additive diffusion; it requires no derivative bound for the trained network.

4.2.4. A post-training certificate

Use the deterministic clipping threshold b_ϕ and expectation-tail allowance τ_ϕ constructed in Appendix E.4. They account for the unbounded Gaussian terminal-dispersion term and are not estimated from the final sample range. Let $X_{\phi,h}^c$ be $X_{\phi,h}$ clipped to $[-b_\phi, b_\phi]$, so that $|\mathbb{E}X_{\phi,h} - \mathbb{E}X_{\phi,h}^c| \leq \tau_\phi$.

For n fresh independent paths, let $\bar{X}_\phi^c$ and s_ϕ^2 be the clipped empirical mean and unbiased variance. Fix a family of at most F one-sided statements before inspecting this noise bank, and set $\ell = \log(2F/\alpha)$. Define

$$L_\phi = \bar{X}_\phi^c - \sqrt{2s_\phi^2 \ell / n} - \frac{14b_\phi \ell}{3(n-1)} - E_{\phi,h}^{\text{all}} - \tau_\phi. \quad (19)$$

In the implementation the empirical mean is enclosed downward and the variance and every subtracted quantity upward. The upper payoff endpoint uses the same terms with reversed signs. The factor F counts both tails when both are reported.

THEOREM 1—Policy-specific continuous-time certificate: *Condition on the training and validation data, and suppose the specified final simulation paths are independent draws from the declared innovation law. Under the arithmetic contract and transfer assumptions above, with probability at least $1 - \alpha$, simultaneously for all declared policies and initial states,*

$$J_y(a^\phi) - J_y(m_0) \geq L_\phi, \quad 0 \leq V^*(y) - J_y(a^\phi) \leq G_d - L_\phi. \quad (20)$$

The optimum V^ still ranges over all adapted feasible controls in the original continuous economy, not merely the networks, schedules, or sampled controllers used in the experiment. Common shocks may be reused across policies; independence across those policies is not required. The guarantee is pointwise in each declared initial state and conditional on training, rather than uniform on an uncountable state domain.*

PROOF: Apply the empirical Bernstein inequality of [Maurer and Pontil \(2009, Theorem 4\)](#) to $(X_{\phi,h}^c + b_\phi)/(2b_\phi)$ and use a union bound over the declared one-sided statements. Rescaling gives the first two error terms in (19). The clipping expectation bound and Proposition S.11 then give the first inequality in (20). Subtract it from the deterministic anchor bound (S.35). Feasibility yields nonnegative regret. An

observed negative upper regret endpoint would indicate failure of the confidence event or of an implementation assumption, not a negative optimality gap. *Q.E.D.*

Unlike $G_d + D_{d,\epsilon}$, the new endpoint depends on the deployed policy through its paired outcomes and sample variance. It can distinguish seeds and unsuccessful policies in the same architectural tube. When $L_\phi > 0$, the post-training guarantee is tighter than the analytical schedule's guarantee. This is a statement about the verified policy, not a deterministic promise that every training run improves welfare. The concentration inequality, the identity that transfers a lower improvement bound into a regret bound, and conservative policy improvement are established ideas (Kakade and Langford, 2002, Pirotta et al., 2013, Scherrer et al., 2015). The contribution here is their operational connection to the continuous capital economy through (18) and the transfer bound (S.44), without a state-covering certificate or a global bound on the trained network's derivatives.

5. BELLMAN EVALUATION AND DIRECT METHOD COMPARISONS

The NBO map has two separate empirical obligations. Its evaluation block must produce a useful continuation-value derivative, and the resulting policy must justify the resources used to construct it. Verification of a policy does not by itself answer the second question. We therefore connect evaluation error to Hamiltonian error and compare candidate-generation methods by a direct paired payoff statistic. The economic problem and the feasible competitor class remain unchanged.

5.1. What the critic contributes

Use the normalized inner product $\langle x, z \rangle_d = d^{-1} x^\top z$ and norm $\|x\|_d^2 = \langle x, x \rangle_d$. At a state of the capital economy, set $q = dD_y V^\pi$. The part of the Hamiltonian depending on consumption is

$$H_q(m) = d^{-1} \sum_i \log m_i - \frac{\zeta}{2} \bar{m}^2 - \langle q, m \rangle_d.$$

On the original action box, and on any convex tube contained in it, H_q is μ -strongly concave for $\mu = \bar{m}_{\text{box}}^{-2}$. Here $\bar{m}_{\text{box}}$ denotes the primitive upper consumption-share

bound, not the cross-sectional mean. Let m_q maximize H_q , let $\hat{q}$ be a critic costate, and let $\tilde{m}$ be within δ of the maximum of $H_{\hat{q}}$ over the same action set.

PROPOSITION 2—Evaluation error and feasible improvement: *Under the preceding conditions,*

$$H_q(m_q) - H_q(\tilde{m}) \leq \left(\sqrt{\delta} + \frac{\|q - \hat{q}\|_d}{\sqrt{2\mu}} \right)^2. \quad (21)$$

For a random rollout costate Z and a square-integrable predictor $g(X)$,

$$\mathbb{E}\|Z - g(X)\|_d^2 = \mathbb{E}\|Z - \mathbb{E}[Z | X]\|_d^2 + \mathbb{E}\|\mathbb{E}[Z | X] - g(X)\|_d^2. \quad (22)$$

These statements do not identify the costate of an Euler rollout with the continuous-time costate; a discretization or evaluation bias must be added when that identification is required.

Equation (21) identifies a reason to regress costates rather than only value levels: Hamiltonian loss is quadratic in a suitably measured costate error when the actor solves its improvement problem. Equation (22) identifies both the variance reduction available from conditional prediction and its regression bias. Neither identity promises that a trained critic reduces mean-square error relative to raw rollout derivatives. The existing comparison therefore includes raw-costate improvement without a learned critic, value-only and costate-only critics, changes in block-update counts, and direct policy optimization. These are substantive alternatives, not alternative labels for NBO. The strong-concavity and conditional-projection arguments are established mathematical principles; their role here is to specify a testable evaluation–improvement mechanism, not to claim a new principle of policy iteration.

The continuous-time performance-difference account in Proposition S.9 integrates evaluation and action errors under the candidate’s occupation law. Its full statement, proof, and alternative residual-to-costate conditions are in the supplement. Training-design prediction errors alone do not meet that account. The current finite-step mechanism below instead states its continuation target exactly and carries its welfare comparison to the continuous economy through the separate payoff-transfer bound.

5.2. Direct paired economic comparisons

Let a and b be two fitted sampled controllers, fixed before the final simulation bank. For each independent final path, compute their schedule-relative statistics $X_{a,h}$ and $X_{b,h}$ with exactly the same initial state and Brownian increments, and form

$$D_{ab,h} = X_{a,h} - X_{b,h}.$$

The common schedule's numerical trajectory cancels. The implementation checks equality of its terminal arrays and the initial-state and innovation hashes. The deterministic transfer allowance is

$$E_{ab,h} = E_{a,h}^{\text{policy}} + E_{b,h}^{\text{policy}} + E_{a,h}^{\text{stat}} + E_{b,h}^{\text{stat}},$$

with outward arithmetic and a separate subtraction-roundoff cushion. In particular, it does not add the schedule transfer error twice. If b_a, b_b are the two clipping thresholds and τ_a, τ_b their expectation-tail allowances, use $b_{ab} = b_a + b_b$ and $\tau_{ab} = \tau_a + \tau_b$.

THEOREM 3—Simultaneous method and population comparisons: *Condition on all fitting and validation data. Suppose the final paths follow the declared independent sampling law and the conditional arithmetic and diffusion-transfer contract of Theorem 1. Apply the empirical Bernstein construction in (19) directly to $D_{ab,h}$, with threshold b_{ab} and allowances $E_{ab,h}, \tau_{ab}$. The resulting endpoints L_{ab}, U_{ab} satisfy*

$$L_{ab} \leq J(a) - J(b) \leq U_{ab}$$

simultaneously with the other declared one-sided statements at their joint confidence level. A difference of two schedule-relative lower endpoints is not used.

The same construction applies to $J_{\mathcal{P}}(a) = \sum_j p_j J_{y_j}(a)$ for a prespecified finite initial-state population $\mathcal{P}$. Draw its profile independently on every final path, average the conditional deterministic transfer bounds with the exact population weights, and use a common clipping threshold and tail bound valid for all profiles. The resulting guarantee is integrated over $\mathcal{P}$; it is not uniform over a continuum of initial states or over unobserved training seeds.

The comparison does not require NBO to dominate a direct method. It makes that empirical question answerable by the quantity actually of interest. A lower endpoint below zero is inconclusive about superiority unless the upper endpoint is also below zero. Certified improvement over the schedule and certified improvement over another trained policy remain different statements. Likewise, a smaller policy-regret upper endpoint need not equal the unknown actual regret.

5.3. A signed Bellman account

Consider the finite held-action economy underlying the candidate-occupation bridge. At date k , let π_k be the reference action, let a_k be the selected candidate action, and write $x_b = z + h(c1 + f(z) - b)$. The true continuation is $S_k(x) = \mathbb{E}F_k(x; \xi)$, where F_k excludes the current reward and follows the reference thereafter. Write $\hat{S}_k$ for the frozen learned continuation. Draw a fresh candidate trajectory, a uniform date $K \in \{0, \dots, N-1\}$, and independent antithetic continuation arrays. Use the same array at the two action endpoints. With $\bar{F}_K$ their average, define

$$\begin{aligned} M &= N\{\ell_K(z, a_K) - \ell_K(z, \pi_K) + \hat{S}_K(x_{a_K}) - \hat{S}_K(x_{\pi_K})\}, \\ C &= N\{\hat{S}_K(x_{a_K}) - \hat{S}_K(x_{\pi_K}) - \bar{F}_K(x_{a_K}) + \bar{F}_K(x_{\pi_K})\}, \\ G &= M - C. \end{aligned}$$

Thus the learned continuation cancels *before* clipping or statistical inference. The cancellation is an identity of the constructed observation, not an approximation requiring a small critic error.

PROPOSITION 4—Signed occupation welfare: *Conditional on the fitted candidate and reference, $\mathbb{E}G$ equals their finite Bellman payoff difference. If L_G is a valid lower confidence endpoint for this expectation, D_{hold} bounds the reference holding deficit, and $E_{a,h}^{\text{all}}$ is the established paired continuous-payoff transfer, then on that event*

$$J(a) - J(m^0) \geq L_G - D_{\text{hold}} - E_{a,h}^{\text{all}}.$$

This conclusion does not require the norm-based continuation-error lower account to be positive.

PROOF: Conditional expectation of the common-noise endpoint difference is the corresponding difference in the true continuation. Summing the one-step advantage over the candidate’s occupation law telescopes to its finite payoff minus the reference finite payoff. Uniform selection of K accounts for the factor N . The holding and paired-transfer bounds then give the displayed inequality. Proposition S.12 and the associated range argument in the supplement supply the complete finite-economy, antithetic, clipping, and arithmetic construction used in the experiment. *Q.E.D.*

The shared observation avoids the loss from estimating M and C separately and subtracting two marginal endpoints. It does not make the critic irrelevant to the *construction* of the candidate: the fitted actor has already used it. Nor does it prove that an alternative constructor, such as cached Raw, would produce a worse candidate. Those comparisons require direct paired method evidence.

6. THE ECONOMIC VALUE OF REUSABLE CONTINUATION

The learned Bellman object can be reused when current economic decisions change but their continuation does not. We examine that property within the capital economy. The finite-period experiment changes the current weight on consumption, the current adjustment cost, or a temporary consumption constraint. Every subsequent primitive, innovation law, terminal preference, and reference policy is common across these decisions. The stored numerical coefficients define the finite economy exactly; the experiment does not identify a quarterly Bellman problem with a continuous-time diffusion without a further approximation argument.

Write the first postdecision mean as $x_a(y) = y + h\{c1 + f(y) - a\}$. For a current task θ , let

$$Q_\theta(y, a) = r_\theta(y, a) + S\{x_a(y)\}, \quad S(x) = \mathbb{E}F(x; \xi),$$

where F contains the current innovation and all subsequent reference rewards, excluding the current reward. The finite query catalog, task catalog, action boxes, and complete distribution over training streams are fixed before confirmation.

PROPOSITION 5—The common continuation object: *If two current tasks differ only in their current reward or current feasible set, their postdecision continuation S is*

identical. A cached continuation observation at the same x and the same innovation array is therefore valid for either task. Cached innovations may also be reused when x changes, but the corresponding state recursion and payoff must then be reevaluated. A cached payoff or derivative at a different x is not an exact observation at the new argument.

PROOF: Conditional on x , every future transition kernel, reward, terminal payoff, and reference action is the same. Iterating these common kernels gives the same law and payoff functional, and hence the same expectation. The last assertion follows because the transition recursion depends on its starting state even when its innovations are fixed. Q.E.D.

This distinction determines the comparison. NBO fits one scalar continuation to common rollout labels and queries its derivative for different current choices. The vector-costate fit, cached Raw actor, task-conditioned direct actor, and cached sample-average optimizer are all allowed their own legitimate reuse. In particular, the Raw competitor does not regenerate an innovation array merely because an optimizer revisits a state. The experiment compares the economic decisions and the work required by these complete procedures. It does not infer that every form of continuation reuse requires a neural scalar critic.

PROPOSITION 6—A fixed-dictionary critic’s bias and variance: *Fix postdecision arguments and positive weights before drawing a new costate-label array. Let q be the vector of true continuation gradients, after multiplication by the square roots of these weights, and let $z = q + \varepsilon$, with $\mathbb{E}\varepsilon = 0$ and finite covariance Σ . Fix differentiable scalar features independently of this label array, and let P be the Euclidean orthogonal projection onto their stacked, weighted gradient columns. The least-squares scalar critic has fitted gradients Pz , and*

$$\mathbb{E}\|Pz - q\|^2 - \mathbb{E}\|z - q\|^2 = \|(I - P)q\|^2 - \text{tr}\{(I - P)\Sigma\}.$$

Thus its risk reduction equals the label variance removed outside the gradient dictionary less its approximation cost. If q belongs to that dictionary, the critic weakly reduces this risk, strictly whenever $\text{tr}\{(I - P)\Sigma\} > 0$. The same fitted scalar function and its labels may be reused for every current task having the common continuation in Proposition 5.

PROOF: Write $Pz - q = -(I - P)q + P\varepsilon$. Orthogonality removes the cross term, and $P = P^\top = P^2$ gives $\mathbb{E}\|P\varepsilon\|^2 = \text{tr}(P\Sigma)$. The raw-label risk is $\text{tr}(\Sigma)$, proving the identity and its consequences. Neither step uses a Gaussian law or independence of the coordinates of a label. Common continuation makes the target gradient and its scalar feature representation independent of the current task; evaluating an already fitted function for another task therefore does not require fitting it again. *Q.E.D.*

The identity isolates a possible statistical contribution of learning the continuation. It also specifies its price: a gradient dictionary can impose enough approximation error to outweigh its variance reduction. The retained random-feature diagnostics and the registered nonlinear comparisons are designed to address this tradeoff; the identity does not determine their empirical signs. Its exact scope is a dictionary fixed independently of the labels used for this least-squares fit and risk at the declared arguments. The fully trained nonlinear NBO critic requires independent economic comparisons rather than identification with that linear projection. A Raw procedure that uses the same projection receives the same risk calculation. Task reuse eliminates repeated fitting of a common object, while evaluation, optimization, and any new state recursions remain part of each method's recorded work.

6.1. When gradient compression improves economic decisions

A quadratic specialization turns the statistical identity into an exact economic comparison. It is useful because it identifies both the potential gain from the common object and the approximation cost that may reverse that gain. It is not used to identify the nonlinear capital experiment with a quadratic model.

Consider current tasks $j = 1, \dots, K$, with positive economic weights ω_j , known b_j , positive-definite H_j , and Bellman objectives

$$Q_j(a; q) = c_j + b_j^\top a - \frac{1}{2}a^\top H_j a - h_j q^\top a.$$

Here $q \in \mathbb{R}^p$ is the common continuation marginal value and $h_j > 0$. Suppose the true optimizer and the optimizers obtained from each estimator below lie in the interior of the specified feasible boxes almost surely. This condition makes the displayed unconstrained first-order solutions feasible; without it the following

equality is not asserted. Let $z = q + \varepsilon$, where $\mathbb{E}\varepsilon = 0$ and $\text{Cov}(\varepsilon) = \Sigma$. Set

$$W = \sum_{j=1}^K \omega_j h_j^2 H_j^{-1}, \quad y = W^{1/2} q, \quad \Omega = W^{1/2} \Sigma W^{1/2}.$$

Fix a scalar-gradient dictionary independently of z . Let P be the Euclidean orthogonal projection onto its columns after multiplication by $W^{1/2}$. The Raw rule uses $\hat{q}_R = z$; the fitted scalar-gradient rule uses $\hat{q}_C = W^{-1/2} P W^{1/2} z$. Each chooses $a_j(\hat{q}) = H_j^{-1}(b_j - h_j \hat{q})$.

PROPOSITION 7—The decision value of the common critic: *Under the preceding conditions,*

$$\begin{aligned} & \sum_{j=1}^K \omega_j \mathbb{E} [Q_j(a_j(\hat{q}_C); q) - Q_j(a_j(\hat{q}_R); q)] \\ &= \frac{1}{2} [\text{tr}\{(I - P)\Omega\} - \|(I - P)y\|^2]. \end{aligned} \tag{23}$$

In particular, if y belongs to the weighted gradient dictionary and $\text{tr}\{(I - P)\Omega\} > 0$, fitting the common scalar object strictly improves expected weighted economic payoff relative to these Raw actions. With misspecification, strict improvement holds exactly when the removed economic-weighted variance exceeds the displayed approximation cost.

PROOF: Completing the square in each true objective gives

$$Q_j(a_j(q); q) - Q_j(a_j(\hat{q}); q) = \frac{1}{2} h_j^2 (\hat{q} - q)^\top H_j^{-1} (\hat{q} - q).$$

Sum with weights ω_j . Raw expected loss is $\frac{1}{2} \text{tr}(\Omega)$. The critic's weighted estimation error is $-(I - P)y + P W^{1/2} \varepsilon$. Its expected squared norm is $\|(I - P)y\|^2 + \text{tr}(P\Omega)$ by orthogonality. Subtracting the losses proves the identity and both strictness statements. Q.E.D.

The comparison is deliberately conditional on a fixed dictionary, a common label law, exact quadratic maximization, and the stated feasibility condition. Adaptive nonlinear feature learning, different actors, boundary constraints, and unequal label budgets require their own analysis. A Raw implementation permitted to apply the

identical projection has the identical estimator and payoff; the proposition does not award a method-name advantage. Its economic content is the value of the estimated common continuation representation under an explicit comparison.

Reuse also has a transparent work threshold. If complete shared construction costs are F_C, F_R , and per-task deployment and optimization costs are c_C, c_R , total work differs by

$$(F_C - F_R) + K(c_C - c_R).$$

When $c_R > c_C$ and $F_C > F_R$, weak work advantage begins at the smallest integer K with $K(c_R - c_C) \geq F_C - F_R$. Validation, loading, action optimization, and any required state recursions belong to these costs. Shared cached labels may reduce F_R or c_R as well. Economic noninferiority cannot be inferred from this accounting identity; it must be established separately on the declared paired payoff event.

Binding task constraints also admit a positive decision-value result. Proposition S.16 in the supplement replaces full interiority by stable active faces and weights continuation errors by the directions in which the tasks can respond. Its strict-gain condition compares removed economically relevant variance with approximation cost; it is not inferred from a favorable fitted-policy endpoint.

6.2. Information and the implemented decision rule

The economic institution is a controller that receives capital observations at specified dates, stores an internal state, and holds the selected consumption share until the next decision date. Its economic primitive is the original capital diffusion. Its internal numerical state need not equal the physical state. When the controller uses private randomization, that randomization is part of its specified information and is included in the law of the evaluated policy.

The exact-history representation in Proposition S.22 assumes continuous noiseless observation of every capital coordinate, known coefficients and actions, exact drift integration, and an independent private Brownian motion. Under those assumptions the observation history supplies a Brownian representation with the law required by the innovation-driven controller. It does not recover the original latent null-space shock path by path. Proposition S.23 states why the added independent

randomization leaves the economic optimum unchanged. These are representation results for a randomized history-dependent policy.

Finite observations require a different implementation account. Proposition S.24 bounds the difference between the measured-state controller and its specified protected parent, retaining sensor error, the internal recurrence, action-map error, clipping, and arithmetic. Its allowance belongs to that parent and that time grid. The empirical comparison reports it alongside the gain measured for the same parent and population; a certificate for a different mesh is not substituted. Complete assumptions and proofs are in Appendix K.

7. ECONOMIC PERFORMANCE AND INDEPENDENT CONFIRMATION

The continuation-learning comparison and the finite-observation exercise address separate questions in the capital economy. The former assesses continuation evaluation and feasible improvement under its stated volatility parameters and complete method distribution. The latter evaluates the original saved policies with a protected parent at their original volatility parameters. Their controller identities, information sets, and probability accounts remain distinct.

7.1. *The capital experiment*

The experiment uses the capital state equation and objective in (11)–(12), in dimensions ten and fifty. It changes the volatility parameters to $\sigma_I = 0.6$ and $\sigma_C = 0.3$; the remaining primitives are $T = 1$, $\rho = 0.04$, $a = 0.10$, $\kappa = 0.15$, $\zeta = 0.2$, $\chi = 0.03$, and $[\underline{m}, M] = [0.02, 2]$. The initial population assigns equal mass to nine profiles with means $-0.5, 0, 0.5$ and cross-sectional spreads $0, 0.25, 0.5$. The methods share the declared radius-0.1 feasible tube, coupling matrix, initial population, and payoff evaluator. This is a parameter regime of the same economic model. The earlier comparison, scalar study, and protected-observation experiment retain their original volatility parameters and their separate evidence records.

Training uses a global grid of 64 cells. Its reference policy is the cell-average analytical schedule and remains fixed through all three work levels. The cumulative occupation samples contain 256, 512, and 1,024 states. Each state comes from an independent reference trajectory at a uniformly selected decision node. Postdecision

inputs are the pre-noise mean and the next decision date. NBO and Raw share four-
 antithetic continuation targets; all member rollouts and all reused labels are counted.
 These sample sizes define increasing information budgets, not equal floating-point
 work across methods.

At each stage NBO fits the spatial continuation derivative by full-replay Sobolev
 mean-square loss, with 1,200 Adam updates followed by L-BFGS. The latter has
 nominal iteration and evaluation limits of 300 and 400; the current line search is
 completed and every actual closure call is counted. A separate time component
 then receives 400 value-fitting updates with the spatial component fixed. NBO and
 Raw use the same feasible actor objective and minibatch/update streams, with 200,
 400, and 800 additional actor updates at the three stages. DPO and HJB use 100, 200,
 and 400 additional updates. The minibatch size is 128; the actor, spatial-critic, time-
 component, and HJB learning rates are 0.003, 0.002, 0.003, and 0.002, respectively.
 Model weights and replay continue across stages. These are declared optimization
 procedures; a common sample budget does not make their optimization costs
 identical.

Online certification and independent final confirmation both use 2,048 decision
 cells and 8,192 paths per stream. The selected fixed networks are deployed on that
 finer grid. The learned 64-cell continuation derivative is consequently a predictor in
 the fine-grid mechanism assessment, whose error account includes that discrepancy;
 it is not identified with an exact fine-grid or continuous-time costate. The protocol
 and numerical source were frozen before the confirmation experiment. Exploratory
 pilots that selected the design remain distinct from the confirmatory observations.

The measured process clock begins before a fresh worker is launched and ends
 after its final evidence is durably written. It includes imports, model setup, de-
 terministic constants, fitting, checks, weight reloading, confirmation, and output.
 Dependency installation and artifact extraction precede this clock and are not
 charged as method execution time.

7.2. Contribution-isolating comparisons

The comparison separates continuation regression, direct policy optimization, and direct approximation of the Bellman equation. The direct HJB method uses the same two learned scalar components as NBO: a time network of widths (16, 16) and a state–time network of widths (32, 32), each with hyperbolic tangent activations and a scalar output. Its known lift is the full-value lift, so the terminal condition is imposed exactly. NBO’s known lift instead represents its postdecision continuation target. This difference follows from the objects being approximated; the learned residual capacity is the same. Both methods use the prespecified occupation sample and the same available model information. No reference solution enters either training objective.

Write $c = a - (\sigma_I^2 + \sigma_C^2)/2$ and $\Sigma\Sigma' = \sigma_I^2 I + \sigma_C^2 \mathbf{1}\mathbf{1}'$. The direct method minimizes the squared residual of

$$\begin{aligned} 0 = & -v_t - \max_{m \in \mathcal{M}_t} \{u(y, m) + (c\mathbf{1} + \kappa \tanh(By) - m)' D_y v\} \\ & - \frac{1}{2} \text{tr}(\Sigma\Sigma' D_y^2 v) + \rho v, \quad v(T, y) = g(y), \end{aligned} \quad (24)$$

where $\mathcal{M}_t$ is the action tube shared by the four primary methods. For $q = dD_y v$, the action-dependent Hamiltonian is

$$H_q(m) = \frac{1}{d} \sum_i \log m_i - \frac{\zeta}{2} \bar{m}^2 - \frac{1}{d} \sum_i q_i m_i. \quad (25)$$

It is strictly concave. Its maximizer is obtained from the scalar consistency equation $z = d^{-1} \sum_i m_i(z)$, with $m_i(z) = \text{proj}_{[\ell_t, u_t]}(q_i + \zeta z)^{-1}$ when $q_i + \zeta z > 0$, and $m_i(z) = u_t$ otherwise. The implementation uses the same 44 monotone bisection steps as the other feasible costate comparisons. The direct HJB policy evaluates a critic gradient and this maximization at every decision. NBO deploys its separately learned actor. The resulting difference in online computation is included in validation, stopping, confirmation, and deployment costs.

Two independent Gaussian probe banks estimate the covariance Hessian trace. If $\hat{R}_1$ and $\hat{R}_2$ are the corresponding residual estimates, then, conditional on a colloca-

tion state and the current critic,

$$\mathbb{E}[\widehat{R}_1 \widehat{R}_2] = R[v]^2. \quad (26)$$

The product, rather than the square of one estimate, is the training loss. Its realized value can be negative. Negative batches are retained. This identity specifies an optimization objective; the common payoff verifier supplies the economic comparison.

Classical scalar training. The original volatility calibration also has a scalar comparison. A monotone implicit Howard method solves the one-state problem and saves an action at every time and state node. These saved policies are deployed by interpolation on fresh diffusion paths, together with the analytical schedule and the hash-pinned scalar NBO and DPO policies from the earlier experiment. Full-box and matched-tube classical policies are separate candidates. Three successively refined state–time grids are followed by an enlargement of the state interval at fixed spatial and temporal spacing. The deployment meshes aggregate the same fine Brownian increments. Policy differences are reported both over the declared initial population and at its individual states -1 , 0 , and 1 . Historical neural fitting clocks retain their historical interpretation; the new classical solves and all new deployment work are measured separately.

The scalar reference has a numerical error account at the level of the declared finite-grid equations. Its algebraic error account uses an outward residual calculation. State-domain approximation and diffusion time discretization remain separate from this algebraic account. The scalar comparison supplies an independent low-dimensional accuracy and work benchmark; the high-dimensional method comparison uses its own payoff data.

7.3. The distribution of a computational method

A computational method includes its initialization, training draws, validation rule, stopping rule, and response to failure. We specify these objects jointly. Let $\mathcal{S}$ be the sixteen complete execution streams listed in the protocol, fixed before the confirmation experiment, and let $\pi_{m,s}$ denote the policy returned by method m under stream s . The randomized implementation draws s uniformly from $\mathcal{S}$;

the experiment executes every member of this distribution. A failed fit returns the feasible analytical schedule and remains in the denominator. Conditional on the stated software environment, the stream specifies the complete training and checking computation. The distribution does not include arbitrary unrecorded initializations or a new draw of the stopping bank on re-execution.

Write $J(\pi)$ for the payoff in the original continuous-time economy, averaged over the specified nine initial-state profiles. The method-level objects are

$$G_m = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \{J(\pi_{m,s}) - J(\bar{\pi})\}, \quad (27)$$

$$\Delta_{m,b} = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \{J(\pi_{m,s}) - J(\pi_{b,s})\}. \quad (28)$$

The first object is improvement on the schedule. The second is the direct comparison of two randomized computational procedures with the same declared stream distribution. Enumeration accounts for the entire training distribution; there is no sampling error from an unobserved subset of these sixteen streams. The remaining uncertainty concerns their independent payoff simulations and the numerical transfer to continuous time. An assertion about this finite distribution is not an assertion about all possible training initializations.

The stopping target is a gain of 0.0005. At each of three declared work checkpoints, fitting pauses for a new, independent certification calculation. Training stops when its lower endpoint attains the target. Failed checks, initialization, action search, checkpoint input and output, and the independent final verification are included in the measured work. The gain 0.001 is a secondary final-attainment threshold; it is not a second measured stopping experiment. A failure to attain either target is retained.

7.4. Simultaneous inference for the finite method distribution

Each returned policy receives N fresh confirmation paths. The confirmation bank is independent of fitting, validation, and every stopping decision. Methods share initial profiles and innovations within a stream, whereas the confirmation banks of distinct streams are independent. A direct contrast is formed on each common

path. Its common analytical schedule cancels, and its transfer account contains the two policy allowances rather than two copies of the schedule allowance, when both entries are simulated candidate gains. An analytical fallback has the exact gain zero. A comparison with that fallback retains the other candidate's full schedule-relative transfer account; two analytical fallbacks have the exact difference zero.

For one registered endpoint, let X_{sj} be the numerical path statistic, $Y_{sj} = \max\{-C_s, \min(C_s, X_{sj})\}$, and let β_s enclose the difference between the continuous-time endpoint θ_s and $\mathbb{E}Y_{sj}$. Thus β_s includes the appropriate diffusion, implementation, clipping-tail, and arithmetic accounts. Put $C = \max_s C_s$, $n = |\mathcal{S}|N$, $\bar{\beta} = |\mathcal{S}|^{-1} \sum_s \beta_s$, and denote the pooled mean and unbiased sample variance by $\bar{Y}$ and $\hat{v}$.

PROPOSITION 8—Finite-distribution comparison: *Conditional on the complete executed policy ensemble, suppose the confirmation observations are independent across pairs (s, j) and the stated range and transfer accounts hold. For $0 < \delta < 1$, the interval*

$$\bar{Y} \pm \left\{ \sqrt{\frac{2\hat{v}\log(4/\delta)}{n}} + \frac{14C\log(4/\delta)}{3(n-1)} + \bar{\beta} \right\} \quad (29)$$

covers $|\mathcal{S}|^{-1} \sum_s \theta_s$ with probability at least $1 - \delta$. The observations need not be identically distributed across streams.

The confirmation family assigns 0.02 across 238 two-sided events: in each dimension, four schedule comparisons and three direct NBO comparisons, each reported for sixteen streams and their uniform method mean. Online stopping receives a separate 0.01 across its 384 possible checks. The mechanism and protected-observation families each receive 0.01. Their union therefore has error probability at most 0.05 for the new experiment. The historical experiment retains its separately stated confidence event. Unused checks do not release probability for additional comparisons.

The practical-equivalence margin is fixed at 10^{-4} payoff units. For a direct interval $[L, U]$, material superiority requires $L > 10^{-4}$, and practical equivalence requires $-10^{-4} < L \leq U < 10^{-4}$. Statistical superiority, $L > 0$, is reported separately. An interval containing zero does not establish equivalence. Under log flow utility, the fixed margin corresponds to a proportional consumption fee $1 - \exp\{-10^{-4}/A\}$,

where $A = (1 - e^{-\rho T})/\rho$; for the stated primitives this is approximately 1.0201 basis points.

Simultaneous stream intervals also enclose the probability that the randomized implementation attains a target g . If they are $[L_s, U_s]$, the corresponding bounds are

$$\frac{\#\{s : L_s \geq g\}}{|\mathcal{S}|} \leq \frac{\#\{s : \theta_s \geq g\}}{|\mathcal{S}|} \leq \frac{\#\{s : U_s \geq g\}}{|\mathcal{S}|}. \quad (30)$$

No binomial sampling interval is attached to a population that has been enumerated in full. Work is likewise reported over all streams, with unattained runs retained in the distribution of work to target. A median over successful runs alone is not the method's median.

The economic decomposition records production, consumption, and terminal dispersion contributions on the same numerical confirmation paths. These components add to the observed numerical payoff gain and help describe the policy's behavior. They have no separate continuous-time confidence statement: the transfer allowance is attached to the total payoff and is not divided among components without additional bounds.

7.5. Method performance, robustness, and complete work

The confirmation record contains all 128 declared method–dimension–stream executions. Tables I and II report the finite-distribution payoff objects defined in Section 7.3. Every endpoint is reconstructed from the original final arrays after checking their common-path identities and source manifests.

Tables III and IV measure the executed stopping procedures, including unsuccessful checks and final confirmation. Table V describes the numerical channels of the payoff change without assigning the total continuous-time certificate to its separate components.

Economic performance of the declared methods. Continuation evaluation and feasible improvement produce positive continuous-economy gains in both dimensions. NBO's method means are approximately 0.001268 and 0.001342 in dimensions ten and fifty; their simultaneous lower endpoints exceed 0.000943 and 0.000957,

TABLE I

PAYOFF OF THE FINITE-DISTRIBUTION COMPUTATIONAL METHODS

d	Method	Mean gain	Lower	Upper	At least target
10	NBO	0.001268	0.000943	0.001592	16/16
10	Raw	0.001267	0.000942	0.001592	16/16
10	Direct policy	0.001262	0.000938	0.001587	16/16
10	Neural HJB	0.000973	0.000647	0.001299	0/16
50	NBO	0.001342	0.000957	0.001727	16/16
50	Raw	0.001343	0.000958	0.001728	16/16
50	Direct policy	0.001347	0.000962	0.001732	16/16
50	Neural HJB	-0.000156	-0.000558	0.000246	0/16

Notes: Means average all sixteen declared execution streams. Endpoints enclose continuous-time payoff gains on the analytical schedule. The last column counts streams whose unrounded simultaneous final lower endpoint attains 0.0005; it is a lower bound on attainment under the declared finite distribution. No stream is excluded. Every lower endpoint is rounded down and every upper endpoint up to six decimal places; displayed intervals therefore contain the original protected intervals. Means are descriptive and rounded to nearest.

TABLE II

DIRECT METHOD COMPARISONS AT THE DECLARED ECONOMIC MARGIN

d	Comparison	Mean	Lower	Upper	Decision
10	NBO–Raw	0.000001	-0.000362	0.000364	Unresolved
10	NBO–Direct policy	0.000005	-0.000358	0.000368	Unresolved
10	NBO–Neural HJB	0.000295	-0.000073	0.000663	Unresolved
50	NBO–Raw	-0.000001	-0.000429	0.000426	Unresolved
50	NBO–Direct policy	-0.000005	-0.000432	0.000423	Unresolved
50	NBO–Neural HJB	0.001498	0.001051	0.001945	Superior

Notes: Pathwise differences use common profiles and innovations within each stream. The equivalence margin is 10^{-4} payoff units. Superior and inferior refer to economic differences exceeding that margin. Decisions use the original unrounded endpoints. An unresolved comparison is not evidence of equivalence. The schedule transfer cancels between two simulated candidates; comparison with an exact analytical fallback retains the other candidate's full schedule-relative transfer. Every lower endpoint is rounded down and every upper endpoint up to six decimal places; displayed intervals therefore contain the original protected intervals. Means are descriptive and rounded to nearest.

TABLE III

COMPLETE MEASURED WORK OF THE STOPPING PROCEDURES

d	Method	Stopped	Mean (s)	Median (s)	Restricted (s)	MiB
10	NBO	0/16	134.82	–	900.00	474.3
10	Raw	0/16	122.67	–	900.00	473.0
10	Direct policy	0/16	285.53	–	900.00	489.4
10	Neural HJB	0/16	373.35	–	900.00	477.3
50	NBO	16/16	357.80	376.03	357.80	530.0
50	Raw	16/16	346.22	365.55	346.22	531.7
50	Direct policy	16/16	596.94	627.10	596.94	545.5
50	Neural HJB	0/16	1690.10	–	900.00	528.4

Stopped counts actual online certificates at the 0.0005 target. Mean is total consumed parent-clock time through independent final confirmation. Median is the lower 0.5 quantile of work to target, with failures assigned infinity; a dash denotes an unattained median. Restricted work is the mean of min(time to target, 900 seconds), retaining failures at 900. Memory is the maximum fresh-worker peak resident size over streams. These clocks describe the recorded execution environment.

TABLE IV

RECORDED OPERATION COUNTS OF THE COMPLETE STOPPING PROCEDURES

d	Method	Train 10^6	Verify 10^6	Actor	Critic	Grad. 10^6	Search 10^6
10	NBO	0.20	134.22	1400	5700	1.04	0.00
10	Raw	0.20	134.22	1400	0	0.00	0.00
10	Direct policy	23.00	134.22	700	0	0.00	0.00
10	Neural HJB	0.07	134.22	0	700	67.20	2956.73
50	NBO	0.18	127.93	1250	5344	0.93	0.00
50	Raw	0.18	127.93	1250	0	0.00	0.00
50	Direct policy	23.00	134.22	700	0	0.00	0.00
50	Neural HJB	0.07	134.22	0	700	67.20	2956.73

Entries are means over all sixteen streams. Train and verify count simulator transitions; actor and critic count optimizer updates; grad. counts first-derivative rows; search counts action-search iterations, including HJB deployment. These are work proxies, not equivalent FLOPs. Full first- and second-derivative, forward, backward, and other counters are retained in the machine-readable report. The parent clock additionally includes every failed operation and all I/O.

TABLE V

DESCRIPTIVE DECOMPOSITION OF THE NUMERICAL PAYOFF GAIN

d	Method	Production	Consumption	Dispersion	Total
10	NBO	0.002303	-0.001271	0.000236	0.001268
10	Raw	0.002303	-0.001273	0.000237	0.001267
10	Direct policy	0.002279	-0.001248	0.000231	0.001262
10	Neural HJB	0.001830	-0.001009	0.000152	0.000973
50	NBO	0.002297	-0.001343	0.000387	0.001342
50	Raw	0.002297	-0.001342	0.000389	0.001343
50	Direct policy	0.002275	-0.001310	0.000382	0.001347
50	Neural HJB	0.000910	-0.001517	0.000451	-0.000156

Entries are observed finite-step common-path contributions, averaged uniformly over the complete stream set. Consumption is minus the consumption deficit; dispersion is the terminal contribution. Their pathwise sum equals the numerical gain. No component continuous-time confidence interval or causal attribution is asserted; the separate total-payoff account remains authoritative.

respectively. These statements average over all sixteen declared complete algorithm streams. All 128 executions returned fitted policies; none used the analytical fallback.

The direct comparison identifies a material advantage over the specified neural HJB procedure in dimension fifty. With the original transfer account, the NBO-minus-HJB interval is contained in $[0.001051, 0.001945]$, whose lower endpoint exceeds the prespecified 10^{-4} economic margin. The original ten-dimensional interval, contained in $[-0.000073, 0.000663]$, is inconclusive. All four direct comparisons with cached Raw and direct policy optimization are also inconclusive under that account. Thus the experiment establishes a domain in which the NBO evaluation-actor construction improves economic performance relative to direct residual fitting with the stated common representation capacity. The finding concerns the declared architectures, training schedules, stopping rule, and finite stream distribution; it is not a ranking of every possible implementation of the three alternatives.

Economic accuracy and complete work. In dimension fifty, all sixteen NBO, Raw, and DPO executions attained the online gain target 0.0005, while no HJB execution did. Their mean complete work was approximately 357.80, 346.22, 596.94, and

1690.10 seconds, respectively, including unsuccessful checks and final confirmation. NBO therefore attained the same declared economic target with less measured work than DPO on the recorded environment; Raw was cheaper than NBO. HJB's unattained runs remain in the work distribution. The reported 900-second restricted-mean functional for an unattained target is a censoring convention, not the actual work consumed or an assertion that the run stopped at 900 seconds.

No ten-dimensional execution attained the online stopping target. Independent final confirmation nevertheless places every returned NBO, Raw, and DPO policy above 0.0005 in that dimension, as it does in dimension fifty. These final certificates do not alter the earlier stopping decisions. No method has a positively certified stream attaining the secondary 0.001 threshold. The complete stream and check-point tables retain this difference between measured stopping and final economic accuracy.

Sources of the observed gain. The numerical decomposition records how the learned decisions change the capital economy. For NBO, production contributes approximately 0.002303 and 0.002297 in dimensions ten and fifty, while the consumption contribution is approximately -0.001271 and -0.001343 . The terminal-dispersion contribution is approximately 0.000236 and 0.000387. Their sums equal the observed numerical gains. These quantities describe the production benefit and the consumption cost of state-dependent decisions on the same paths; the continuous-time confidence statement belongs to the total payoff, not separately to these components.

7.6. Direct comparisons with a shared diffusion-transfer account

The direct statistic already compares policies on common innovations. The deterministic transfer can use that same coupling. Proposition S.17 in the supplement bounds the error of the payoff difference for two held policies directly, retaining the original decision grid and realized actions. The fitted policies, simulated observations, stopping decisions, clipping, tail account, and simultaneous confidence family remain unchanged.

The following table reports the original and refined endpoints together. The refinement concerns a proved deterministic allowance; it is not an additional simulation experiment or a retrospectively changed stopping rule. Comparisons involving a schedule fallback retain their stated fallback account. The refined comparison

TABLE VI
ORIGINAL AND REFINED DIRECT METHOD ENDPOINTS ON THE SAME OBSERVATIONS

d	Comparison	Mean	Original interval	Refined interval	Decision
10	NBO–Raw	0.000001	[-0.000362, 0.000364]	[-0.000110, 0.000112]	Unresolved
10	NBO–Direct policy	0.000005	[-0.000358, 0.000368]	[-0.000106, 0.000117]	Unresolved
10	NBO–Neural HJB	0.000295	[-0.000073, 0.000663]	[0.000179, 0.000411]	Superior
50	NBO–Raw	-0.000001	[-0.000429, 0.000426]	[-0.000133, 0.000130]	Unresolved
50	NBO–Direct policy	-0.000005	[-0.000432, 0.000423]	[-0.000137, 0.000127]	Unresolved
50	NBO–Neural HJB	0.001498	[0.001051, 0.001945]	[0.001347, 0.001650]	Superior

The new deterministic common-innovation transfer is proved in Appendix H. Both intervals use the identical original clipped statistic, path identities, empirical Bernstein event, and confidence allocation. The refinement was developed after the original source freeze and inspection of the first transfer width.

Only the deterministic transfer changes; mixed analytical fallbacks retain the original allowance. Decisions use the original 10^{-4} economic margin. Original training, stopping, and work records remain unchanged.

establishes economically material NBO gains over the specified neural HJB method in both dimensions. Its direct lower endpoints exceed 0.000179 and 0.001347, respectively, and hence exceed the original 10^{-4} margin. The ten-dimensional comparison changes from unresolved under the original account to materially positive under the proved refinement; the fifty-dimensional conclusion is strengthened. The four NBO–Raw/DPO intervals remain unresolved: they contain zero and extend outside the practical-equivalence band. They also do not certify noninferiority at the stated margin. The conclusion follows from a sharper deterministic account of the same paired observations, with the original statistical and computational records retained.

7.7. Fresh confirmation of the paired comparison

The paired-transfer refinement in the preceding table was developed after the original numerical source was fixed. It is a deterministic refinement on the original event, but that chronology alone does not make it a prospectively executed calculation. We therefore preserve the original and refined old-bank intervals and evaluate every one of the 128 original selected policies on an independent bank after fixing the paired theorem, its constants, the interval code, and the new confidence allocation.

Each of the sixteen original streams receives 8,192 fresh paths on 2,048 time cells. There is no retraining, checkpoint replacement, or alteration of historical online stopping. The new 0.01 familywise allocation covers its 238 declared events; it is not borrowed from the earlier confidence family. Table VII gives all direct method comparisons.

TABLE VII

PROSPECTIVELY FIXED PAIRED TRANSFER ON FRESH CONFIRMATION BANKS

d	NBO minus	Mean	Original transfer	Fixed paired transfer
10	Raw	0.000001	[-0.000365, 0.000367]	[-0.000114, 0.000115]
10	Direct policy	0.000005	[-0.000361, 0.000372]	[-0.000109, 0.000120]
10	Neural HJB	0.000295	[-0.000076, 0.000667]	[0.000176, 0.000415]
50	Raw	-0.000001	[-0.000432, 0.000429]	[-0.000136, 0.000133]
50	Direct policy	-0.000005	[-0.000436, 0.000426]	[-0.000140, 0.000130]
50	Neural HJB	0.001497	[0.001046, 0.001948]	[0.001342, 0.001653]

All 128 original R15 selected policies are evaluated without retraining on new R16 banks. Each of the sixteen original streams receives 8192 independent paths on 2048 cells. The paired theorem, constants and new familywise allocation $\alpha = 0.01$ precede these observations. Interval endpoints and regret upper bounds are rounded outward. This is independent confirmation of fixed policies; historical stopping is unchanged.

With the prospectively fixed paired transfer, the fresh NBO-minus-HJB intervals are contained in $[0.000176, 0.000415]$ and $[0.001342, 0.001653]$. Both exceed the declared 10^{-4} economic margin at their lower endpoint. The comparison remains with

the original specified HJB implementation. Its purpose is independent confirmation of the fixed policies and the paired calculation, not a claim about a newly trained HJB family or a new economic calibration.

All four fresh NBO-minus-Raw and NBO-minus-direct-policy intervals include zero and extend beyond the equivalence band. Thus neither equivalence nor non-inferiority follows. The fresh broad NBO regret upper bounds are 0.025971 and 0.036845, rounded upward; they are explicit absolute guarantees rather than 10^{-4} near-optimality certificates. The full method table records every method, including the negative fifty-dimensional HJB sample mean.

TABLE VIII
INDEPENDENT R16 CONFIRMATION OF THE FIXED R15 POLICIES

d	Method	Mean gain	Simultaneous interval	Regret bound
10	NBO	0.001269	[0.000943, 0.001596]	0.025971
10	Raw	0.001268	[0.000942, 0.001595]	0.025972
10	Direct policy	0.001264	[0.000937, 0.001590]	0.025977
10	Neural HJB	0.000974	[0.000646, 0.001302]	0.026268
50	NBO	0.001342	[0.000955, 0.001729]	0.036845
50	Raw	0.001343	[0.000956, 0.001730]	0.036843
50	Direct policy	0.001347	[0.000960, 0.001734]	0.036840
50	Neural HJB	-0.000155	[-0.000560, 0.000249]	0.038359

All 128 original R15 selected policies are evaluated without retraining on new R16 banks. Each of the sixteen original streams receives 8192 independent paths on 2048 cells. The paired theorem, constants and new familywise allocation $\alpha = 0.01$ precede these observations. Interval endpoints and regret upper bounds are rounded outward. This is independent confirmation of fixed policies; historical stopping is unchanged.

7.8. Absolute accuracy in the original control problem

Let ν be the declared uniform distribution over the nine initial profiles, and let J_ν^0 denote the analytical schedule's payoff. The pointwise anchor account in (S.35) implies $V_\nu^* - J_\nu^0 \leq B_{d,\nu}$ after averaging over this initial population. The repeated

worker records concern this same bound. We verify their profiles and component weights and retain the maximum of the already averaged upper bounds if their final rounding differs.

For a method m , write $\bar{J}_{d,m} = 16^{-1} \sum_s J_v(\pi_{m,s})$. On the simultaneous confirmation event, its existing gain lower endpoint satisfies $\bar{J}_{d,m} - J_v^0 \geq L_{d,m}$. Consequently,

$$0 \leq V_v^* - \bar{J}_{d,m} \leq B_{d,v} - L_{d,m}. \quad (31)$$

The optimum in this expression ranges over the original adapted action-box class. The returned methods use the declared common tube, which is a feasible subset. The inequality therefore accounts for absolute loss against the original economic problem. It does not identify the unobserved loss or require another probability allocation.

TABLE IX

FULL-CLASS PAYOFF-GAP UPPER BOUNDS AND THE ECONOMIC MARGIN

d	Method	Gain lower	Regret upper	Margin	Ratio
10	NBO	0.000943	0.025971	0.0001	259.8
10	Raw	0.000942	0.025971	0.0001	259.8
10	Direct policy	0.000938	0.025976	0.0001	259.8
10	Neural HJB	0.000647	0.026266	0.0001	262.7
50	NBO	0.000957	0.036843	0.0001	368.5
50	Raw	0.000958	0.036841	0.0001	368.5
50	Direct policy	0.000962	0.036838	0.0001	368.4
50	Neural HJB	-0.000558	0.038358	0.0001	383.6

The regret upper bound is $B_{d,v} - L_{d,m}$, rounded outward: $B_{d,v}$ bounds the analytical schedule's loss against the original full adapted control class, and $L_{d,m}$ is the existing lower endpoint for the uniform mean over all sixteen returned policies. The ratio divides the regret upper bound by the prespecified 10^{-4} payoff margin. All 128 source-bound confirmation records are audited. This deterministic translation uses the existing confirmation event and no additional alpha.

The reported full-class upper bounds range from 0.025971 to 0.038358 payoff units. Their ratios to the declared margin range from 259.8 to 383.6. Thus the current

absolute account gives guarantees at those explicit tolerances. It does not certify proximity to the full optimum within the 10^{-4} margin, even when a relative method comparison is resolved.

Table IX puts this absolute account next to the economic margin used for direct method comparisons. A positive gain lower endpoint establishes improvement on the schedule; a direct comparison resolves a difference between two trained methods; and a small right-hand side of (31) would establish proximity to the full optimum. These implications concern different quantities. The neural HJB comparator supplies an independent candidate construction, while the scalar calculation supplies a separate finite-grid accuracy account. Neither supplies an exact high-dimensional reference value.

7.9. Continuation accuracy where the policy changes the allocation

The mechanism assessment follows the policy that each registered NBO stream actually returns. It draws fresh occupation paths and uniform action bridges after all policy selection. The reference remains the cell-average analytical schedule on one global assessment grid. The saved scalar critic may have been fitted on a coarser grid; its assessment error includes this discrepancy.

Table X compares the fitted costate with a direct four-path antithetic bank on the identical continuation queries. The calculation uses independent cross-products to remove reference-bank variance from the critic's risk estimand. It also reports the paired difference in risk, retaining negative sample contributions and all registered streams. In dimension 10, the costate-risk contrast has endpoints -0.0344447 and 0.0343934; this interval does not determine which predictor has smaller risk.

In dimension 50, the costate-risk contrast has endpoints -0.0225684 and 0.0225961; this interval does not determine which predictor has smaller risk.

Table XI carries the same continuation account through the feasible actor's actual action change and the quantitative continuous-payoff transfer of Theorem S.6. The approximate continuation's scalar advantage is evaluated at both action endpoints. The actor's gap is bounded from the saved global curvature certificate, including the endpoint calculation when strong concavity is unavailable. The combined

Bellman-bridge lower endpoint in dimension 10 is -0.006778. The retained costate, sampling, and implementation allowances do not establish improvement through this sufficient bound.

The combined Bellman-bridge lower endpoint in dimension 50 is -0.004800. The retained costate, sampling, and implementation allowances do not establish improvement through this sufficient bound.

The complete six-statistic intervals and numerical cushions appear in Table O.I. Table O.II reports the additional fresh-process assessment work. These calculations distinguish accuracy of the evaluation stage, quality of the feasible action update, and the independent payoff comparison between complete algorithms.

TABLE X

COSTATE PREDICTION ON THE CANDIDATE'S OCCUPATION LAW

d	Critic risk mean	Raw risk mean	Contrast lower	Contrast upper
10	0.045	0.302	-344.447	343.934
50	0.398	0.259	-225.684	225.961

Notes: All displayed risk entries are multiplied by 10^4 . Each dimension pools 256 independent action bridges from every one of the 16 registered NBO streams, or 4,096 bridges. Each bridge has two independent four-path antithetic continuation banks on the 2,048-cell global grid. The signed critic statistic removes reference Monte Carlo variance in expectation. The raw statistic is the risk of one four-path bank on the identical represented queries. The paired contrast estimates critic risk minus raw risk. Its endpoints use the registered family of 12 two-sided statements with total error probability 0.01 and include interval sample arithmetic and proved clipping tails. These are evaluation-subproblem comparisons; the returned Raw policy's welfare is assessed separately.

7.10. A prospective signed occupation account

The new assessment evaluates every original selected NBO policy on fresh occupation paths. Its proof, clipping interval, confidence family, and 1,024 bridges per stream were fixed before the new bank was drawn. The paired continuation calculation uses the same innovations at the candidate and reference action endpoints. It evaluates the signed continuation correction and applies Proposition S.12 directly to their joint Bellman gain, rather than subtracting separately estimated endpoints.

TABLE XI

FROM THE CONTINUATION ACCOUNT TO ECONOMIC GAIN

d	L_M	$\sqrt{U_A U_E}$	U_D	Holding/transfer	Gain lower
10	-0.000849	0.005630	0.004113	0.000300	-0.006778
50	-0.000153	0.004285	0.002962	0.000363	-0.004800

Notes: The costate upper bound includes the exact-state representation allowance. The holding and transfer column sums the outward holding deficit and the quantitative paired payoff transfer to the original continuous economy. The last column is the larger of the product bound and the optional quadratic actor-gap bound when its positive modulus is verified. A negative endpoint records the limit of this sufficient mechanism account. Actual method welfare remains assessed by the independent direct payoff experiment; the mechanism assessment cannot change selection or stopping.

TABLE XII

A FRESH SIGNED BELLMAN MECHANISM ACCOUNT

d	M mean	C mean	L_G	Hold/transfer	Economy lower
10	0.001243	-0.000002	0.001067	0.000300	0.000767
50	0.001354	0.000001	0.001138	0.000363	0.000776

Notes: Each dimension fully enumerates the original sixteen selected NBO streams, with 1,024 independent occupation bridges per stream. One bank of two antithetic pairs supplies the common-noise continuation endpoints. The fitted scalar continuation cancels in $G = M - C$ before inference. All six two-sided (d, M, C, G) statements share a new familywise error probability of 0.01. The finite lower endpoint includes numerical sample and prefix-representation allowances. The final column additionally retains the original holding deficit and full continuous-payoff transfer. This result does not rank the critic against Raw.

In dimension 10, the signed occupation account certifies a continuous-economy gain of at least 0.000767.

In dimension 50, the signed occupation account certifies a continuous-economy gain of at least 0.000776.

The original negative Cauchy–Schwarz mechanism bounds remain part of the record. The present calculation changes the sufficient error account and uses an independently frozen confirmation bank. Its positive endpoints, when obtained, concern the occupation Bellman identity of the returned policy. They do not establish lower prediction risk than Raw or identify a unique incremental contribution of

the learned critic. The separate common-data method comparison addresses that question.

7.11. Independent scalar training and numerical accuracy

The separate scalar study uses the original volatility calibration, $(\sigma_L, \sigma_C) = (0.15, 0.10)$, and the same one-state capital economy. Eight independent Howard solves generate full-box and matched-tube policies on all four prespecified grids. The policies are then evaluated by actual interpolation-based deployment on 8,192 fresh paths at both 1,024 and 2,048 time steps. The analytical schedule and both frozen scalar neural policies use the same initial states and Brownian paths.

TABLE XIII

INDEPENDENT SCALAR TRAINING AND ALGEBRAIC ACCURACY

Policy	$N_y/N_t/L$	$v(-1)$	$v(0)$	$v(1)$	E_0	Seconds
F1	401/256/6	-3.393994	-1.314658	0.816406	1.08E-12	3.76
F2	801/512/6	-3.393648	-1.314316	0.817061	2.28E-12	9.30
F3	1601/1024/6	-3.393474	-1.314143	0.817386	5.93E-12	24.54
F4	2401/1024/9	-3.393474	-1.314143	0.817386	8.94E-12	30.26
T1	401/256/6	-3.393994	-1.314658	0.816406	1.08E-12	3.77
T2	801/512/6	-3.393648	-1.314316	0.817061	2.28E-12	9.33
T3	1601/1024/6	-3.393474	-1.314143	0.817386	5.93E-12	24.64
T4	2401/1024/9	-3.393474	-1.314143	0.817386	8.94E-12	30.50

F1–F4 solve the full primitive action box; T1–T4 solve the common schedule tube of radius 0.1. E_0 is the outward finite-grid algebraic error bound in Proposition 5.26. The displayed error endpoints are rounded upward; the full outward endpoints are retained in the artifact. Seconds include each new solve, its interval residual audit, and policy serialization. The last grid enlarges the domain from $[-6, 6]$ to $[-9, 9]$ at the same declared spatial and temporal spacing. Boundary data are the stated far-state approximation.

All 8 classical solves reached the stated stopping criterion, using at most 3 Howard updates at each date. Their finite-grid algebraic bounds range from 1.072×10^{-12} to 8.938×10^{-12} . The last joint state–time refinement changes the three initial-state

values by amounts between 1.724×10^{-4} and 3.250×10^{-4} . The subsequent enlargement of the domain changes these values by at most 4.441×10^{-16} . Thus the algebraic solve, the training-grid approximation, and the finite-domain approximation have separate, observable numerical accounts. The small domain change does not bound the remaining time or spatial discretization error.

TABLE XIV

DEPLOYED SCALAR POLICY GAINS ON COMMON DIFFUSION PATHS

Policy	$y = -1$	$y = 0$	$y = 1$	Population	MC s.e.
Schedule	0.0000	0.0000	0.0000	0.0000	0.0000
F1	0.1137	1.5586	0.6768	0.7818	0.0069
F2	0.1137	1.5587	0.6771	0.7819	0.0069
F3	0.1136	1.5586	0.6772	0.7819	0.0069
F4	0.1136	1.5586	0.6772	0.7819	0.0069
T1	0.1137	1.5586	0.6768	0.7818	0.0069
T2	0.1137	1.5587	0.6771	0.7819	0.0069
T3	0.1136	1.5586	0.6772	0.7819	0.0069
T4	0.1136	1.5586	0.6772	0.7819	0.0069
NBO (R12)	0.1036	1.5557	0.5683	0.7412	0.0071
DPO (R12)	0.0468	1.4460	0.6303	0.7065	0.0066

All payoff gains and standard errors are multiplied by 10^3 . The comparator is the analytical schedule evaluated on the same numerical mesh and Brownian path. Population denotes the mean over independent uniform initial-state draws; the last column is its paired Monte Carlo standard error. Initial-state columns retain their separate rankings. NBO and DPO are the prespecified frozen R12 scalar policies. These are conditional Euler-deployment diagnostics; Monte Carlo standard errors do not include continuous-time or boundary approximation error.

At 2,048 deployment steps, the largest-domain classical policy has a mean schedule gain of 7.819×10^{-4} , compared with 7.412×10^{-4} for NBO and 7.065×10^{-4} for DPO. The paired classical advantages are 4.069×10^{-5} (Monte Carlo standard error 6.841×10^{-7}) against NBO and 7.541×10^{-5} (4.063×10^{-7}) against DPO. Both advantages are positive at each of the three reported initial states. The numerical comparison therefore identifies an independently computed feasible classical com-

petitor with higher observed payoffs than both frozen neural policies in this scalar economy.

The neural ordering varies across initial states. NBO minus DPO has population mean 3.472×10^{-5} , but its conditional mean at $y = 1$ is -6.206×10^{-5} . The same reversal occurs on the 1,024-step deployment mesh. This state-specific outcome is retained alongside the population comparison. Corresponding full-box and tube policies have identical payoffs path by path on both reported deployment meshes. The maximum recorded nodal deviation from the schedule is 0.046600, below the tube radius. This observation concerns the computed scalar policies and does not establish that the tube restriction is harmless in every dimension.

The complete full-box and matched-tube value arrays coincide bit for bit on each of the four grids. Because the action sets are nested and the terminal and boundary data are identical, Corollary S.27 converts the two algebraic enclosures into a bound on the loss from the tube restriction in the exact finite-grid problem. Across the four grids, this loss at every initial-date node is at most 1.788×10^{-11} . This deterministic conclusion concerns the stated finite-grid problem; the domain and diffusion approximation accounts retain their separate scope.

The complete scalar process consumed 195.90 seconds, including fresh-process setup, all eight solves, interval audits, serialization, both common-path deployments, and output. The individual new classical solves consumed 3.76–30.50 seconds. Within the fine-mesh deployment, the measured action-evaluation totals were 1.26–1.36 seconds for the classical tables and 9.38–9.60 seconds for the frozen neural maps. These are operation-specific measurements on the recorded worker. The historical neural fitting clocks are not included in a new end-to-end training comparison.

The algebraic error bound concerns the exact equations with their declared boundary data. The independent domain enlargement and paired deployment-mesh comparisons quantify distinct numerical sensitivities. The complete tables retain all trained policies, both meshes, all initial-state strata, and all 110 pairwise numerical comparisons. They supply a scalar accuracy and classical-work benchmark without identifying an exact value for the high-dimensional diffusion.

7.12. *Implementation from finite state measurements*

The implementation question has a direct economic interpretation. A manager observes a vector of log capital stocks at specified dates, computes consumption rates with the saved network, and holds those rates until the next observation. The manager's input consists of the measured physical state. The finite-observation controller does not require a Brownian coordinate or the intervening drift integral. Proposition S.24 applies to this controller through a protected parent with the same saved weights and inward action guard.

The experiment verifies that parent afresh at 1,024 and 4,096 decision dates in dimensions 10 and 50. It retains the original volatility calibration and the previously fixed iteration-80 NBO weights; no outcome in the new observation experiment selects a policy. Table XV subtracts each deterministic sensing allowance from the corresponding parent guarantee. All four parents have positive lower endpoints. Six of the twelve finite-observation lower endpoints remain positive. With 4,096 dates and exact measurements, the continuous-economy gain is at least 0.001050 in dimension 10 and 0.001054 in dimension 50. In dimension 10, a measurement error of 0.0001 retains a lower gain of 0.000889 at the finer grid.

The dependence on measurement precision is economically material. At dimension 50, an error of 0.0001 already exceeds the available parent margin under the current global network bound, and an error of 0.001 removes a positive guarantee in both dimensions. These endpoints give an implementation requirement for the specific measured-state rule. They do not infer the sign of its actual gain when the allowance is too large. The full intervals, allowance-to-gain ratios, and fresh-process costs appear in Tables Q.V and Q.VI.

7.13. *The economic return to reusable continuation*

We next compare repeated current decisions with a common continuation, the object identified in Section 6. Temporary interventions vary current utility weights, adjustment costs, and action constraints; future primitives and the reference policy remain fixed. This is a finite-period capital economy with exactly specified stored coefficients, not a discretization result inferred without continuous-time transfer. Four

TABLE XV
CONTINUOUS-ECONOMY GAIN BOUNDS WITH FINITE OBSERVATIONS

d	N	Parent lower	$\nu = 0$	$\nu = 0.0001$	$\nu = 0.001$
10	1024	0.000766	0.000687	0.000526	-0.000925
10	4096	0.001070	0.001050	0.000889	-0.000559
50	1024	0.000807	0.000252	-0.001038	-0.012651
50	4096	0.001193	0.001054	-0.000239	-0.011873

Notes: All lower endpoints are printed downward to six decimal places. The last three columns are lower bounds for the finite-observation controller’s payoff gain over the analytical schedule. Each parent is freshly verified at the same decision grid as its sensor transfer, using 8,192 independent paths. The four parent intervals share error probability 0.01; the transfers are deterministic. The fixed weights and original volatility calibration are retained from the reviewed study. A negative endpoint means that this error account does not certify improvement at that observation precision.

designs comprise the original low-volatility anchor, the disclosed quarterly-reuse pilot, an unpiloted longer continuation, and untouched intermediate primitives. All are reported in both dimensions.

The five methods are scalar NBO, a fitted vector costate, a materialized Raw actor, a task-conditioned direct actor, and sample-average optimization with its complete innovation cache. All retain legitimate reusable data across three predetermined work stages. Each comparison averages every one of sixteen fixed streams and 384 fixed current queries. Each query has 512 independent antithetic future pairs, shared across methods; an opposite-sign pair is one observation. The 216-event family has its own probability budget 0.018. No stage, seed, or economic design is removed after confirmation.

Table XVI displays the predetermined third stage; Appendix R.5 retains all three. The long fifty-dimensional case gives NBO-minus-vector, NBO-minus-Raw-actor, and NBO-minus-direct-actor intervals contained in $[16.38, 18.13]$, $[18.02, 19.72]$, and $[3.27, 4.25]$, respectively, in units of 10^{-4} . The untouched intermediate fifty-dimensional case also gives material advantages over the Raw and direct actors. These are direct economic comparisons, not an inference from a small critic residual. They concern the specified methods and complete finite menu; they do not assert that every Raw procedure must be worse.

TABLE XVI

INCREMENTAL VALUE OF THE REUSABLE NBO CONTINUATION

Economy	d	NBO – Vector	NBO – Raw actor	NBO – DPO	NBO – SAA
Original	10	[0.00, 0.01]	[0.06, 0.07]	[0.06, 0.07]	[-0.01, 0.01]
Original	50	[0.03, 0.05]	[0.25, 0.26]	[0.24, 0.25]	[-0.03, -0.02]
Quarterly	10	[-0.02, 0.02]	[0.20, 0.26]	[0.12, 0.17]	[-0.06, -0.01]
Quarterly	50	[0.22, 0.27]	[0.82, 0.90]	[0.76, 0.83]	[-0.41, -0.36]
Long	10	[0.14, 0.92]	[5.67, 6.70]	[-1.03, -0.19]	[-1.99, -1.22]
Long	50	[16.38, 18.13]	[18.02, 19.72]	[3.27, 4.25]	[-2.68, -1.89]
Intermediate	10	[-0.01, 0.03]	[0.25, 0.33]	[0.15, 0.21]	[-0.03, 0.02]
Intermediate	50	[0.66, 0.79]	[1.47, 1.63]	[1.30, 1.44]	[-0.49, -0.39]

Notes: Intervals are in units of 10^{-4} and use the final, third work stage. Every cell averages all sixteen declared streams and all 384 fixed current-period queries. Each query has 512 independent antithetic future pairs; opposite signs form one observation. Five methods share each pair. All three stages and all eight economic-dimension cells contribute to one 216-event family with error probability 0.018. Current reward and a known terminal quadratic are removed before the random residual is bounded. The mean column is the clipped residual estimator plus this known offset. Numerical arithmetic, clipping and the unclipped Gaussian law are included. The materiality margin is 10^{-4} .

Cached sample-average optimization is a particularly informative alternative. In six of the eight final-stage cases, the entire direct interval lies within $[-10^{-4}, 10^{-4}]$, establishing practical equivalence at the fixed margin. This is stronger than merely observing an interval that contains zero. In the two long-continuation cases the NBO-minus-sample-average intervals are instead contained in $[-1.99, -1.22]$ and $[-2.68, -1.89]$, in units of 10^{-4} : the sample-average method has a material payoff advantage. The ten-dimensional long case also favors the direct actor in sign, without resolving the 10^{-4} noninferiority question.

The work comparison completes, rather than replaces, the payoff comparison. NBO uses less complete third-stage construction and query work than cached sample-average optimization in every case; this gives a work saving within the equivalence band in the six cases just identified. For example, the quarterly and untouched intermediate fifty-dimensional cases use 14.34 versus 62.90 seconds and 12.70 versus 79.46 seconds per stream. The Raw actor is cheaper than NBO, so the material payoff comparisons do not imply simultaneous dominance in both

TABLE XVII
COMPLETE CONSTRUCTION AND QUERY WORK

Economy	d	NBO	Vector	Raw actor	DPO	Cached SAA
Original	10	12.52	8.73	6.86	16.21	15.60
Original	50	12.67	9.85	7.07	20.24	27.68
Quarterly	10	10.36	7.69	6.09	21.85	25.23
Quarterly	50	14.34	11.15	8.28	38.93	62.90
Long	10	12.05	8.87	6.96	26.53	30.42
Long	50	14.59	11.06	8.16	38.03	67.04
Intermediate	10	11.96	9.16	7.01	36.55	44.71
Intermediate	50	12.70	9.62	7.18	45.58	79.46

Notes: Seconds per stream, averaged over all sixteen streams, including unsuccessful fits. Each entry is the actual outer-process clock for all three stages, imports, source checks, reusable training data, optimizer and derivative work, all saved queries and durable output. Fixed dependency installation and source extraction precede the clock. Independent confirmation is a shared measured bundle, reported separately in the complete account; the table does not allocate one-fifth of that bundle as an observed standalone verification cost.

coordinates. The independent confirmation and scalar audit constitute a shared measured bundle, retained in full and charged conservatively in the complete account. No unobserved standalone verification cost is manufactured by dividing that bundle among methods. Earlier-stage work is a saved-prefix account with disclosed overhead, not an executed economic stopping time.

Absolute accuracy is separately assessed on the prespecified continuous scalar action segment at one fixed high-dimensional state. The independent 0.002 family retains all 128 statements, each based on 65,536 antithetic pairs. Eighty candidates meet the 10^{-4} whole-segment margin: all streams in both original and quarterly dimensions and in the intermediate ten-dimensional case. Forty-eight do not. The upward-rounded largest bounds include 0.0000141 for quarterly dimension fifty, 0.0002846 for intermediate dimension fifty, and 0.1963168 for the long fifty-dimensional case. Table R.IV retains every case. These are conditional accuracy guarantees for the declared scalar class, not full-vector or adapted-control near-optimality statements.

7.14. *A strengthened HJB comparison across economic designs*

A separate experiment addresses the original HJB implementation and economic-variation concerns. The baseline uses exact covariance Hessian contraction, four pre-specified collocation/learning-rate configurations, Adam followed by L-BFGS, and residual-only selection on independent banks. Both safeguarded-Newton greedy deployment and its distilled actor are retained. No final payoff selects between them. Appendix R gives the full construction and charges all shared prerequisites.

The four designs are the historical high and low volatility cases, a longer horizon, and a joint economic stress. Each dimension uses eight new complete training streams, producing 64 trials and 320 policies. Independent confirmation uses 8,192 paths per method on 2,048 cells, with 648 simultaneous events and budget 0.01. The NBO gain lower endpoint is positive in every economic-dimension case. Table XVIII shows, however, that strengthening the comparator changes the comparative conclusion. Only the low-volatility fifty-dimensional NBO-minus-HJB contrasts exceed the materiality margin at their lower endpoints. The other fourteen direct HJB intervals contain zero. The eight NBO-minus-Raw comparisons also remain unresolved, and Raw uses less measured work in each case. The complete method, target-attainment, residual, and cost records appear in Appendix R.4. These results establish feasible improvement over the analytical schedule under genuine economic variation while sharply separating it from a universal method ranking.

8. CONCLUSION

Neural Bellman Operators separate the learning of continuation values from feasible economic improvement. Their useful object is a continuation that is shared across decisions, not a small residual considered in isolation. The evaluation and policy-error results specify how approximation enters decisions; the capital construction places their consequences in economic payoff units.

The signed occupation account provides a positive continuous-economy assessment of the returned NBO policies where the original norm-based sufficient bound was negative. Fresh prospective confirmation also supports the specified NBO-HJB payoff comparison. Neither result, by itself, identifies a unique incremental contribution of the learned critic relative to cached Raw. The latter question is made

TABLE XVIII

NBO RELATIVE TO BOTH STRENGTHENED HJB DEPLOYMENTS

Economy	d	NBO minus greedy HJB	NBO minus distilled HJB
High volatility	10	$[-0.000173, 0.000189]$	$[-0.000171, 0.000191]$
High volatility	50	$[-0.000221, 0.000180]$	$[-0.000161, 0.000240]$
Low volatility	10	$[-0.000129, 0.000177]$	$[-0.000130, 0.000177]$
Low volatility	50	$[0.000334, 0.000697]$	$[0.000340, 0.000704]$
Longer horizon	10	$[-0.000846, 0.000965]$	$[-0.000834, 0.000977]$
Longer horizon	50	$[-0.000897, 0.001423]$	$[-0.000836, 0.001483]$
Joint stress	10	$[-0.001020, 0.001062]$	$[-0.001003, 0.001079]$
Joint stress	50	$[-0.001277, 0.001229]$	$[-0.001051, 0.001455]$

Notes: Outward-rounded continuous-payoff intervals. Every cell includes all eight new training streams. The two-sided family has 648 events and probability budget 0.01. Candidate and deployment selection do not use final payoffs. The materiality margin is 10^{-4} . These are fixed-envelope comparisons, not historical online stopping results.

explicit by the common-continuation, bias–variance, and economic decision-value identities. They give a constructive positive mechanism and its exact approximation price, rather than an unconditional ranking of method names.

The numerical record retains the unresolved Raw and direct-policy comparisons, Raw’s lower original work, broad high-dimensional regret bounds, and classical scalar and state-specific reversals. The completed strengthened-HJB comparison shows why the comparator specification matters: most of the enlarged-family payoff contrasts remain unresolved despite positive NBO gains. The five-method continuation experiment gives a more discriminating account of reuse. NBO has material advantages over fitted alternatives in the difficult long fifty-dimensional case and achieves practical payoff equivalence with cached sample-average optimization at lower construction and query work in six of eight final-stage cases. The sample-average method nevertheless has material payoff advantages in both long-continuation cases. All stages and all economic designs remain visible. Eighty of 128 conditional scalar candidates meet the declared accuracy margin over their entire action segments; the other forty-eight do not, and no full-vector conclusion is substituted for these conditional statements.

The original recursive-utility, preference-formation, temporal-self, and game applications remain part of the theory under their stated assumptions. The current supplement retains their complete derivations. Together, these results make the NBO program a study of learned continuation, feasible decisions, and their economic value, with policy verification and comparative method evidence serving distinct roles.

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